

Mock Exam

March 2024

T. Breezie

Grades 13:1

Chemistry

Montego Bay Community College



Umbrelitics Analysis Report



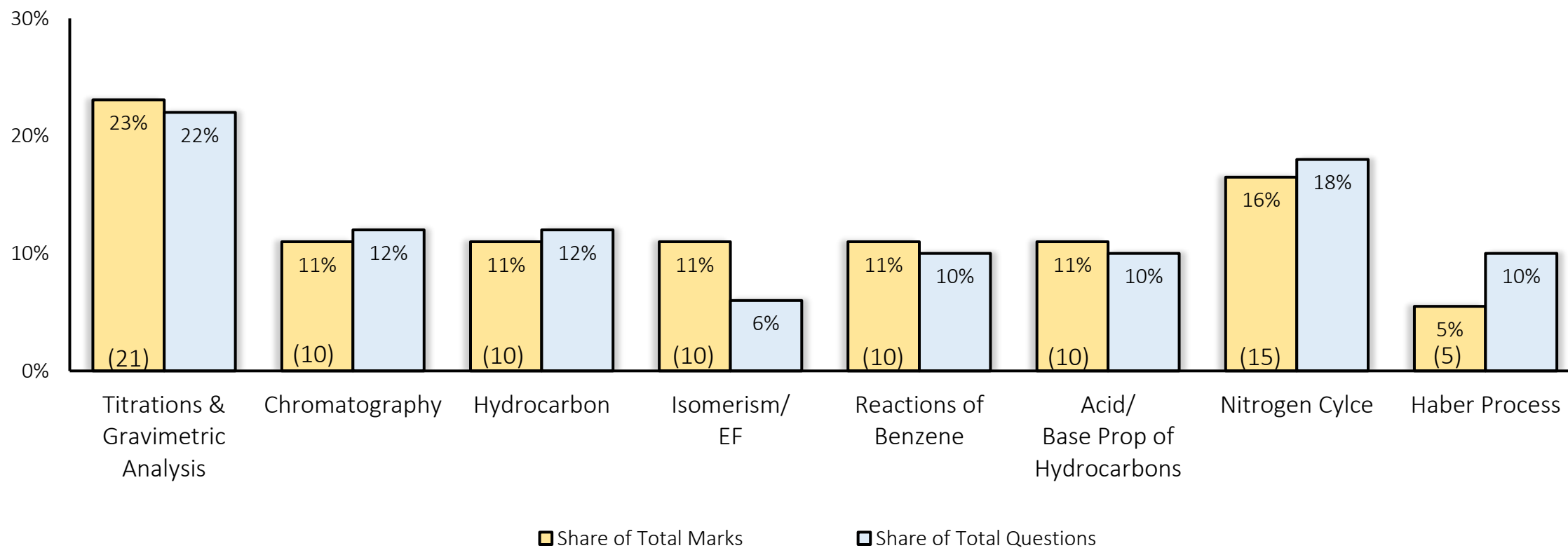


Exam Analysis

- The topic Titration had the largest share of the exam questions, 22%
- The topic Isomerism had 11% of the marks while occupying 6% of the exam questions
- Haber Process occupied 10% of the exam questions, while accounting for 5% of total marks
- Isomerism occupied the smallest portion of the exam, 6%

Total Marks = 91

Share of Total Marks vs Question Count



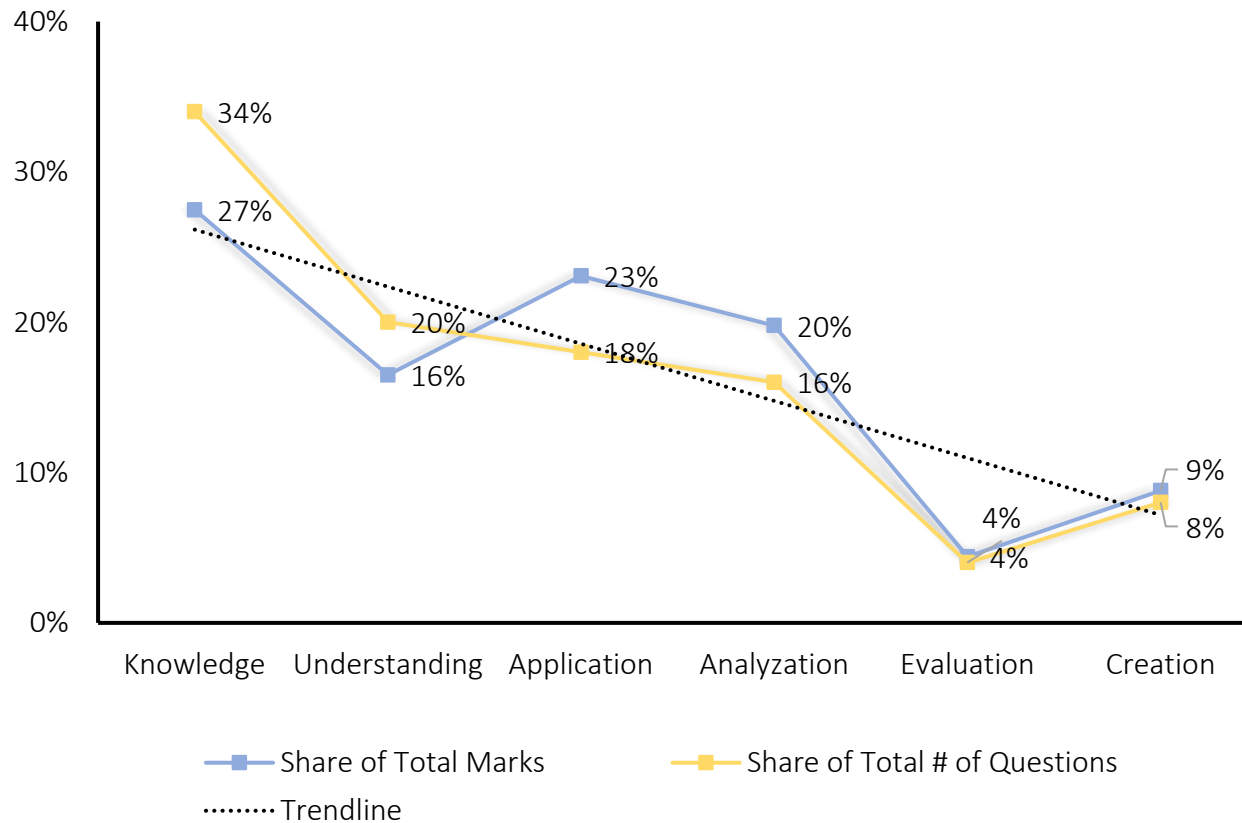
(*) - Total marks allotted to each topic are in parentheses

Exam Structure by Taxonomy & Question Category



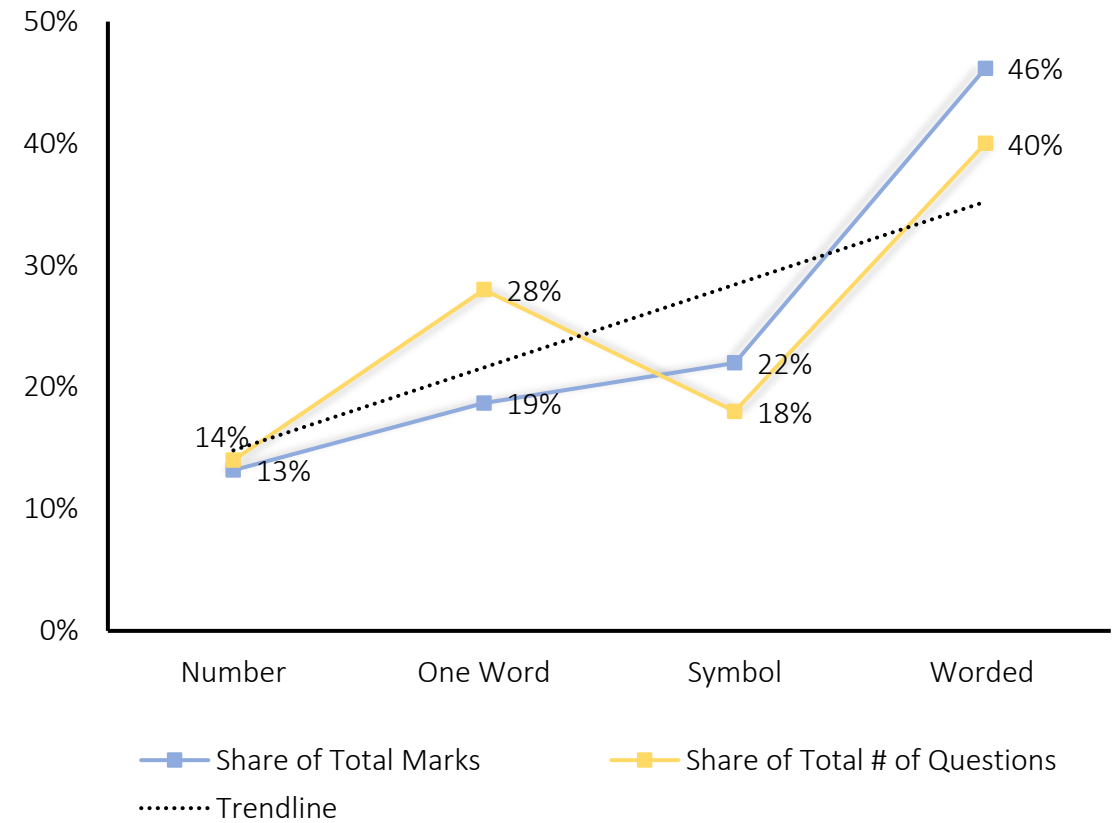
- Knowledge type questions occupied 34% of the total count of questions, while accounting for 27% of total marks
- Application type questions had 23% of the total mark allotment, while occupying 18% of the exam

% Allocation for each Taxonomy



- Worded responses occupied 40% of the exam, while accounting for 46% of total marks

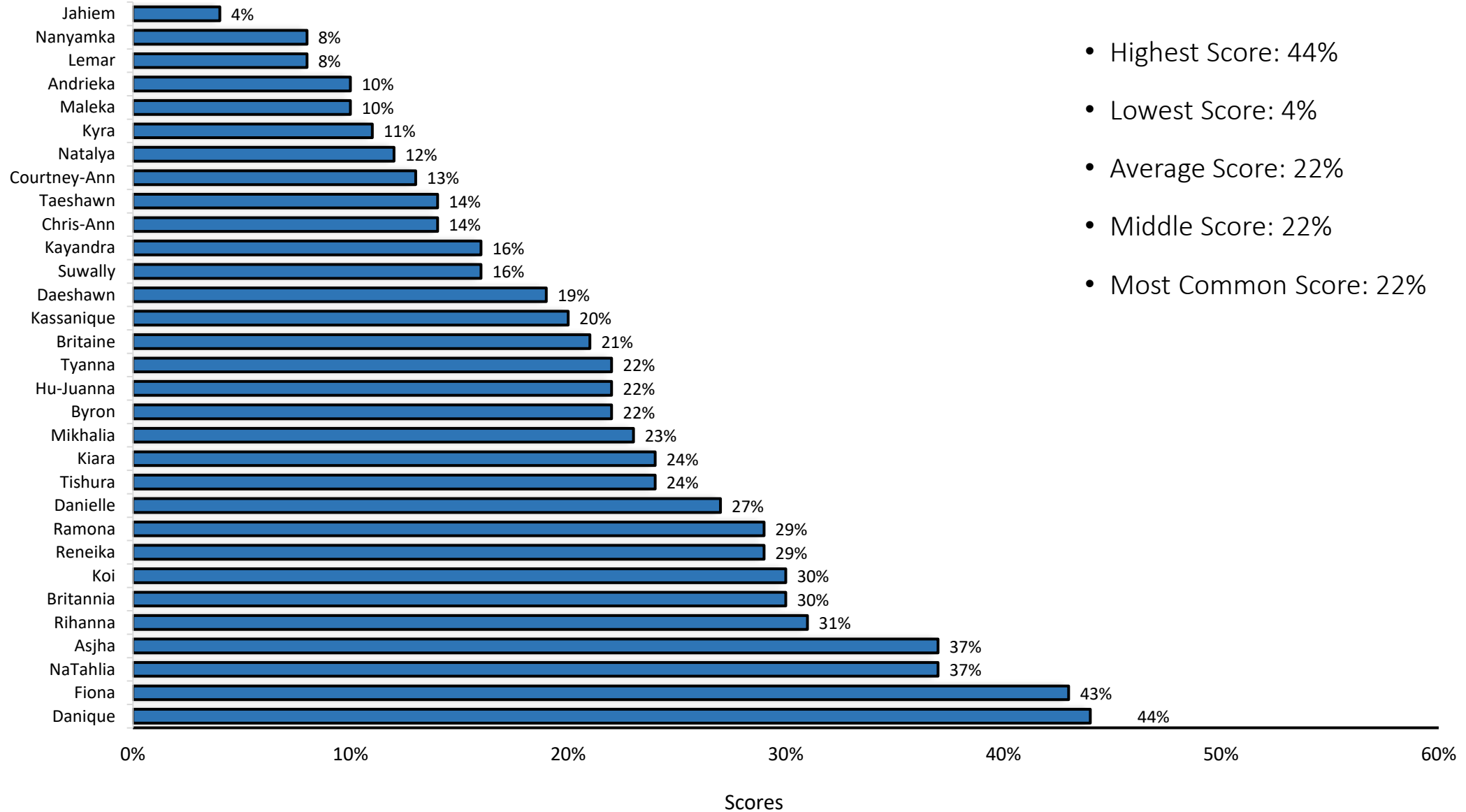
% Allocation for each Response Type





Student Performance Analysis

Grades Distribution

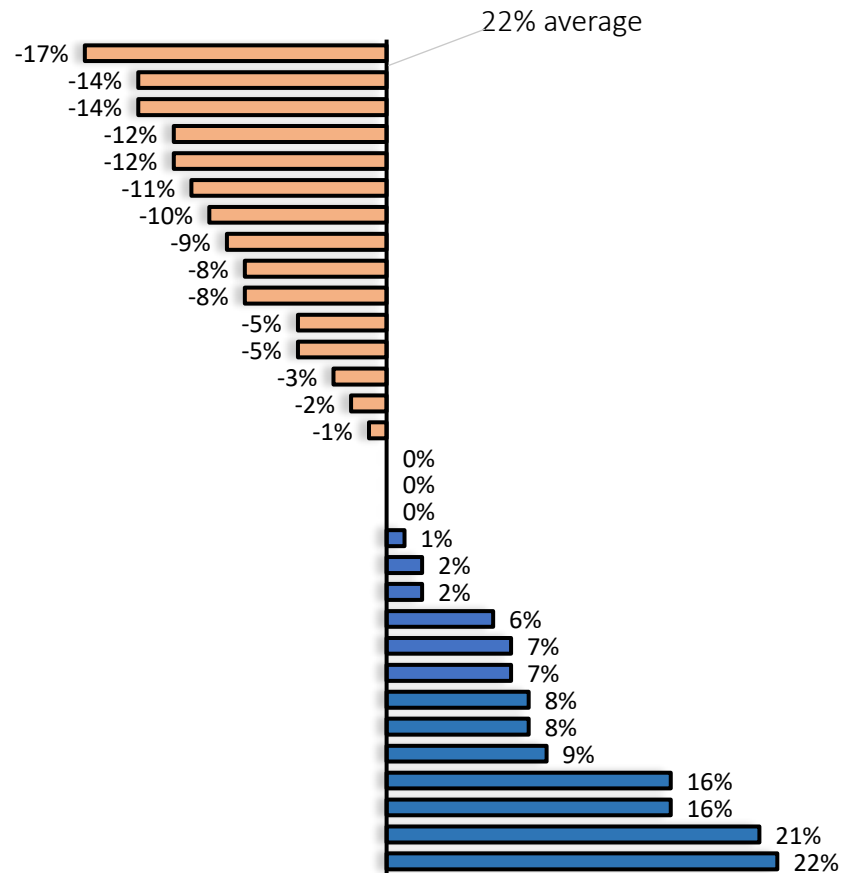




Grade Variance & Distribution

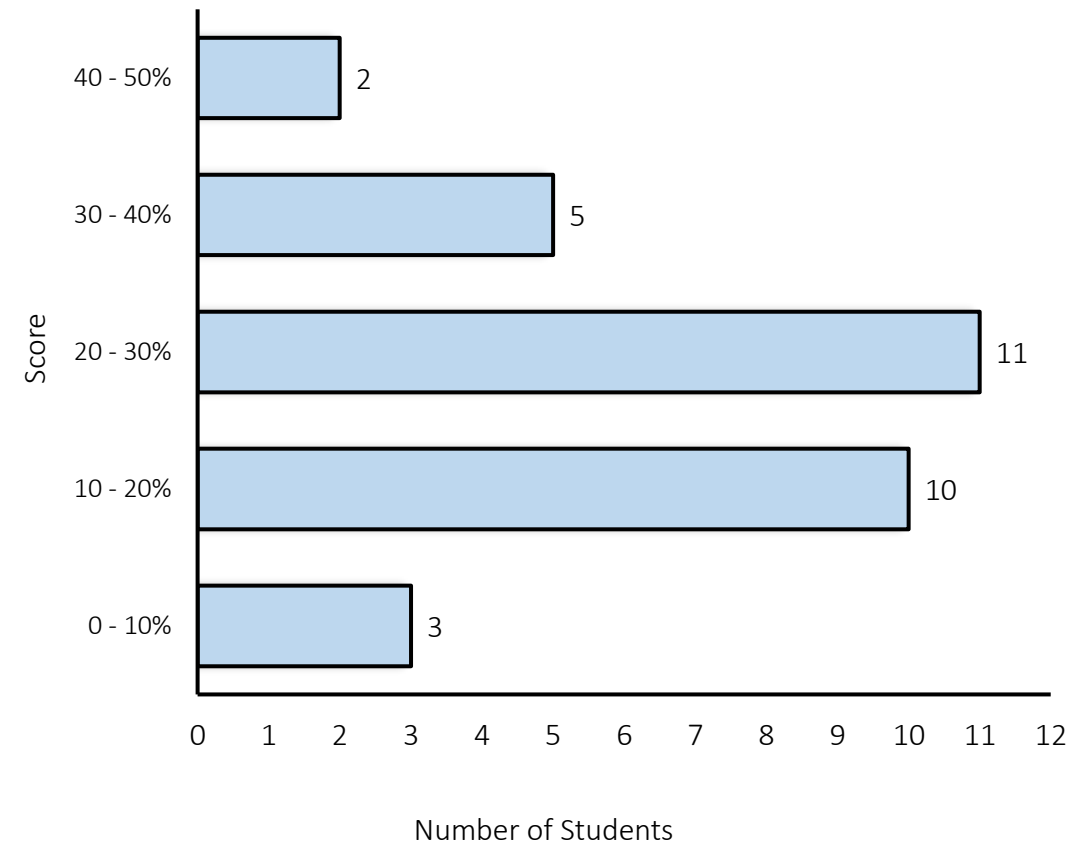
- 16 students scored above or equal to the average of 22%, while 15 scored below

Grade Variance around Average Score



- The most common grade received is between 20 and 30 percent

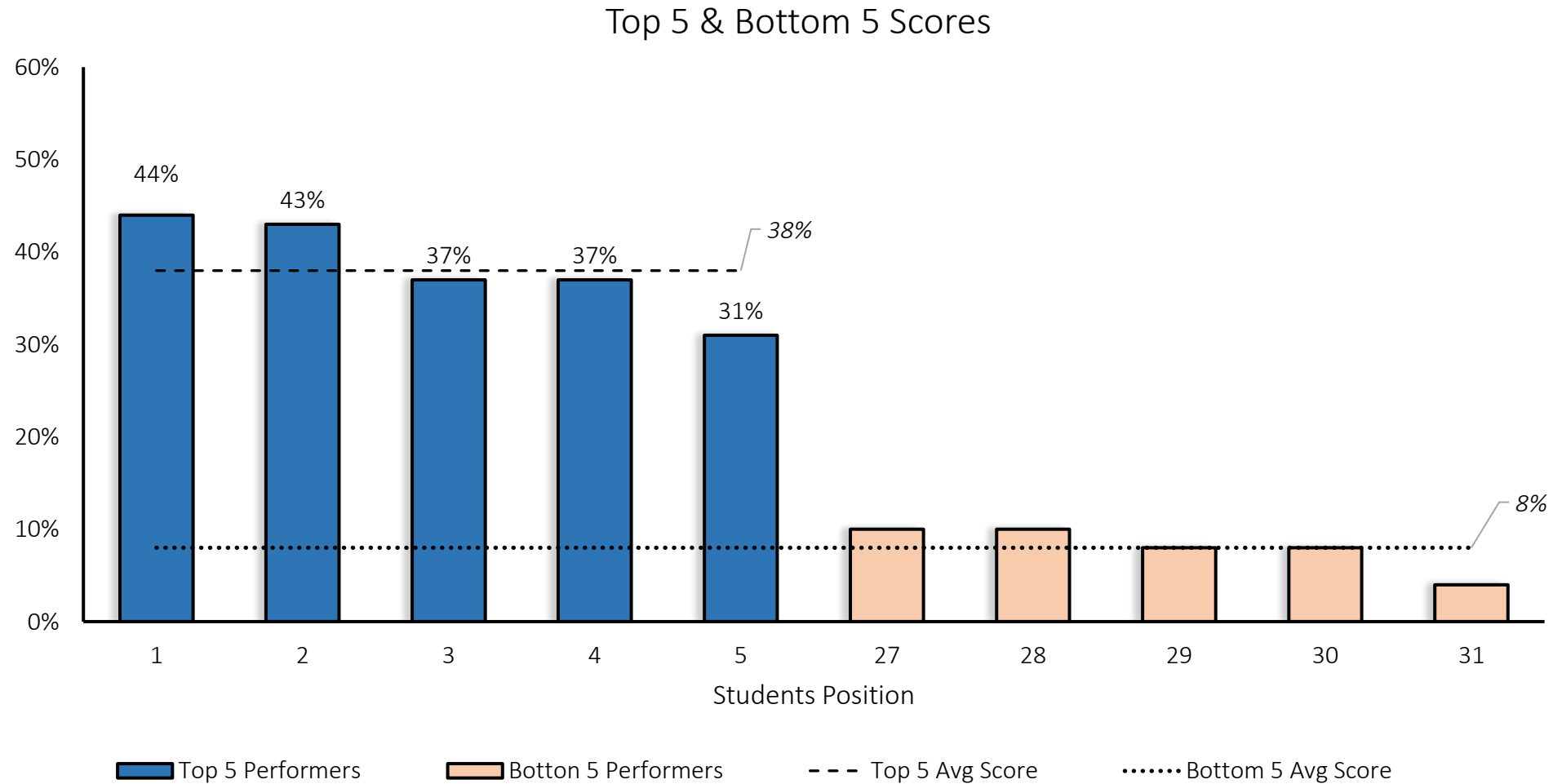
Class Distribution of Scores





Overall Performance by Top 5 & Bottom 5 Scores

- Top 5 performers averaged 38% on the exam, while the bottom 5 performers averaged 8%





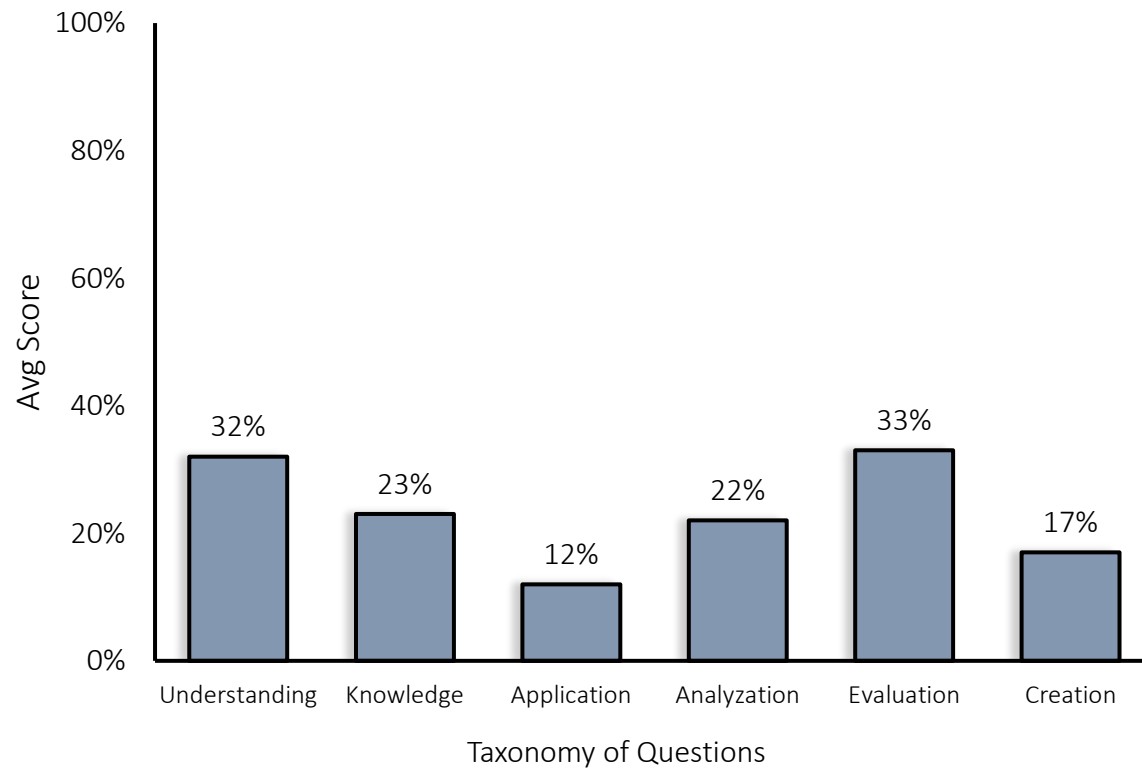
Performance by Taxonomy



Overall Performance by Taxonomy

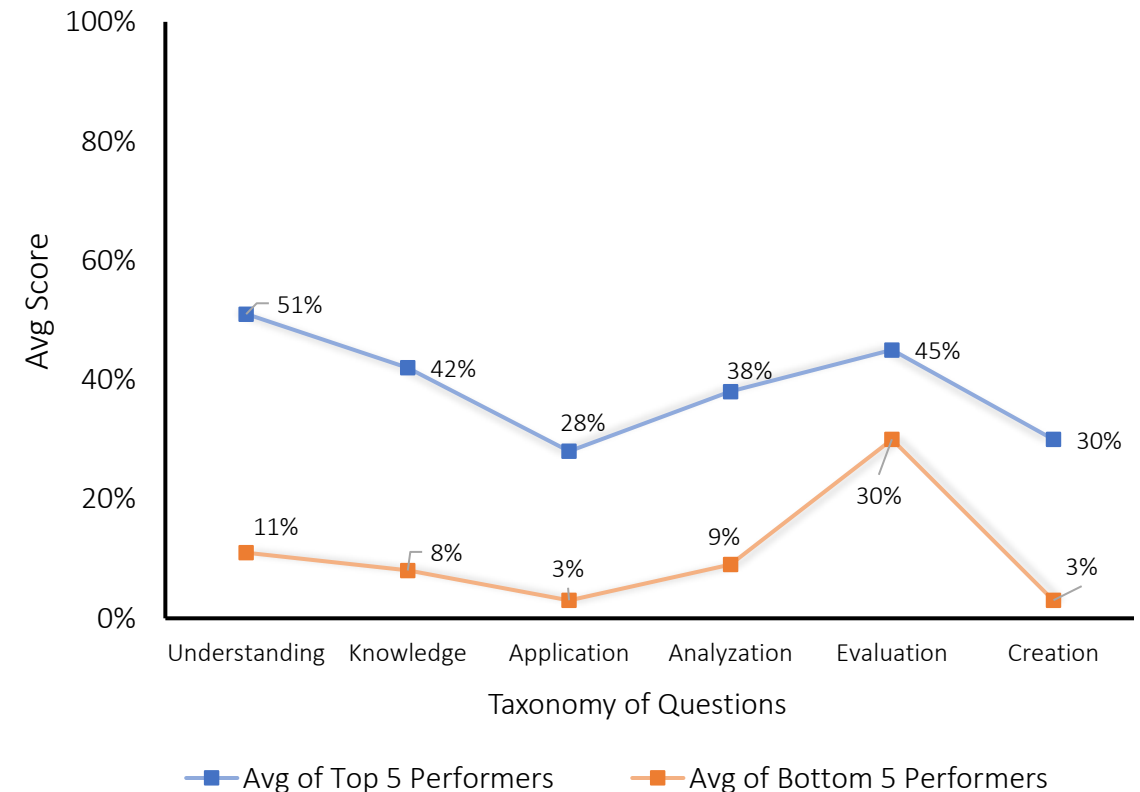
- The Evaluation Taxonomy was the highest performing cognitive skill on the exam, while Application was the lowest performing cognitive skill

Avg Performance for each Taxonomy



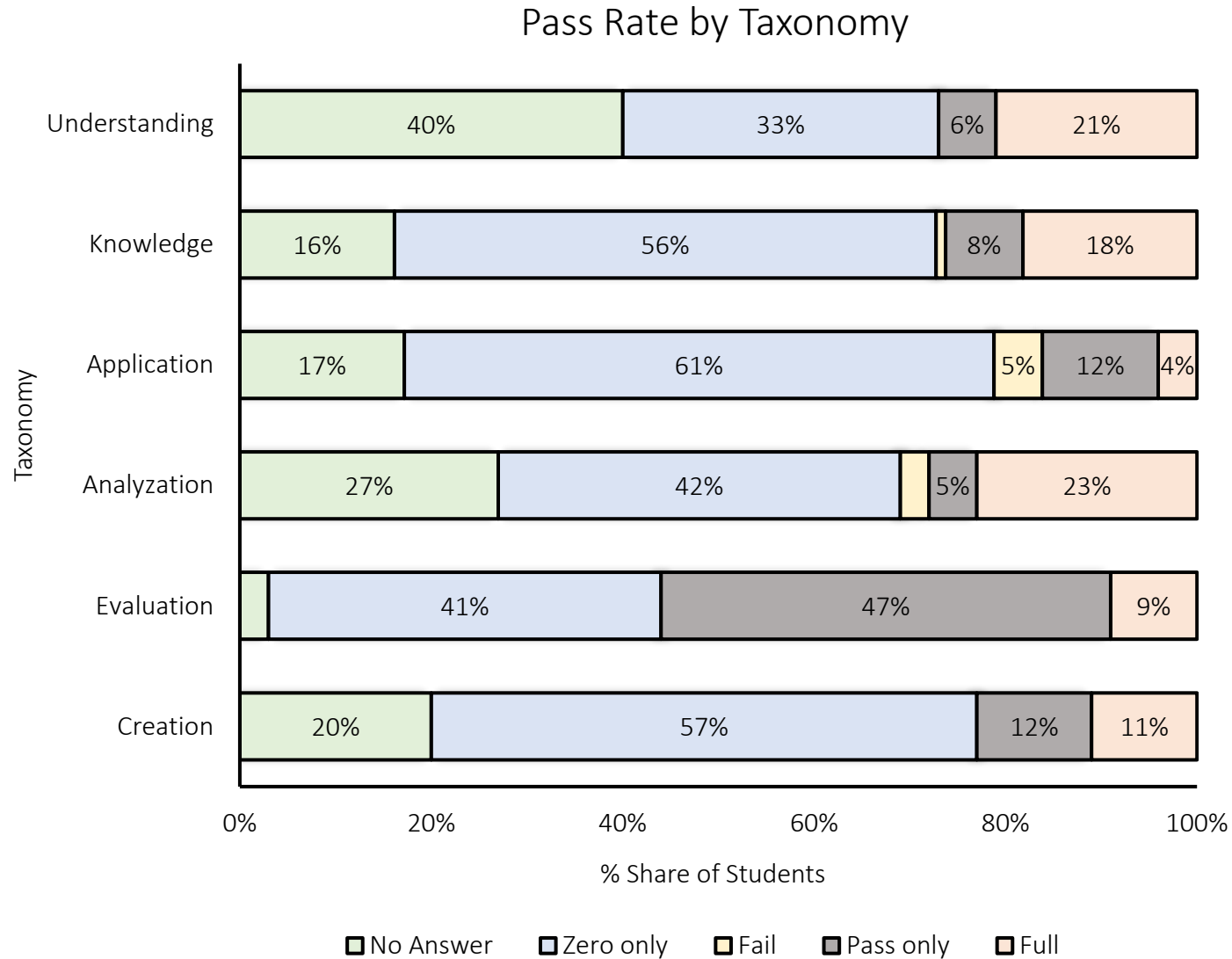
- The Taxonomy Understanding had the widest disparity between top and bottom 5 performers, while the Evaluation cognitive skills had the lowest difference

Avg for Top 5 & Bottom 5 Performance by Taxonomy





Pass Rate by Taxonomy



- Analyze experienced the highest pass rate, while Application received the highest share of zeros

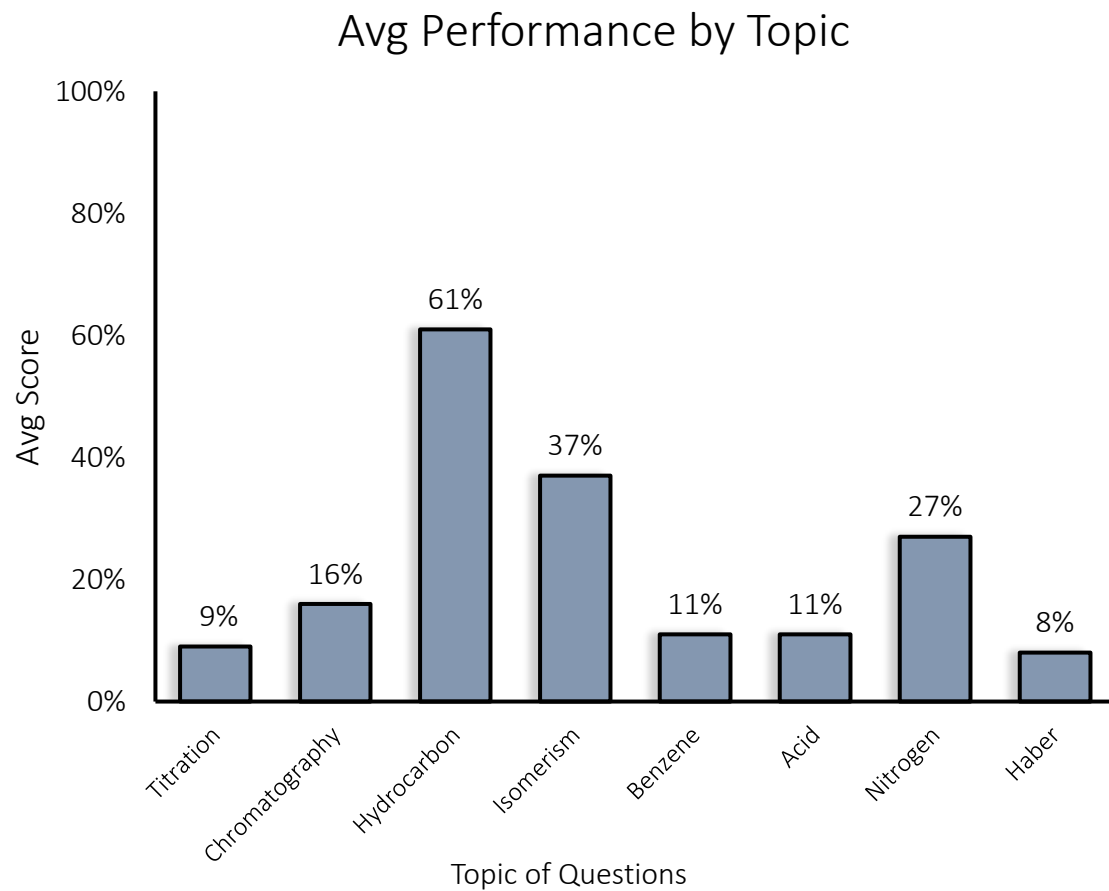


Performance Analysis by Topic



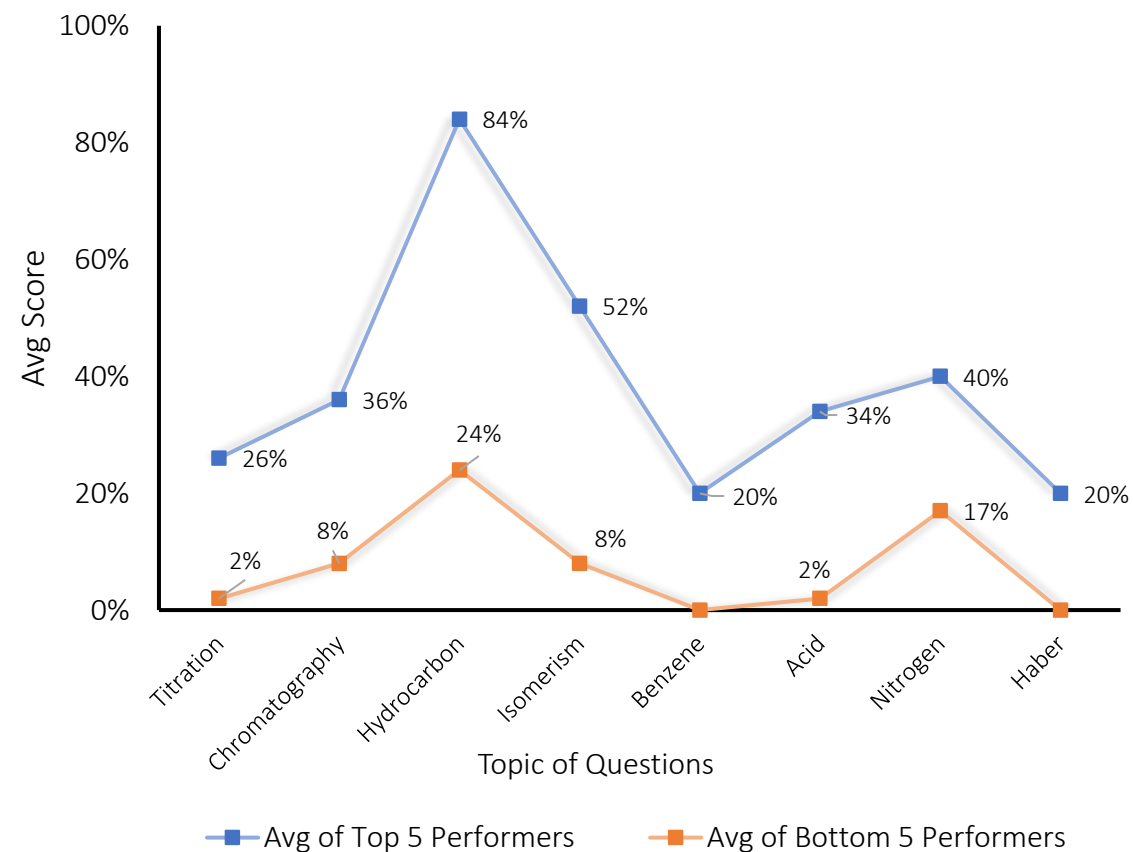
Performance by Topic

- Hydrocarbon structure had the highest average score at 61%
- Haber Process had the highest average score at 8%



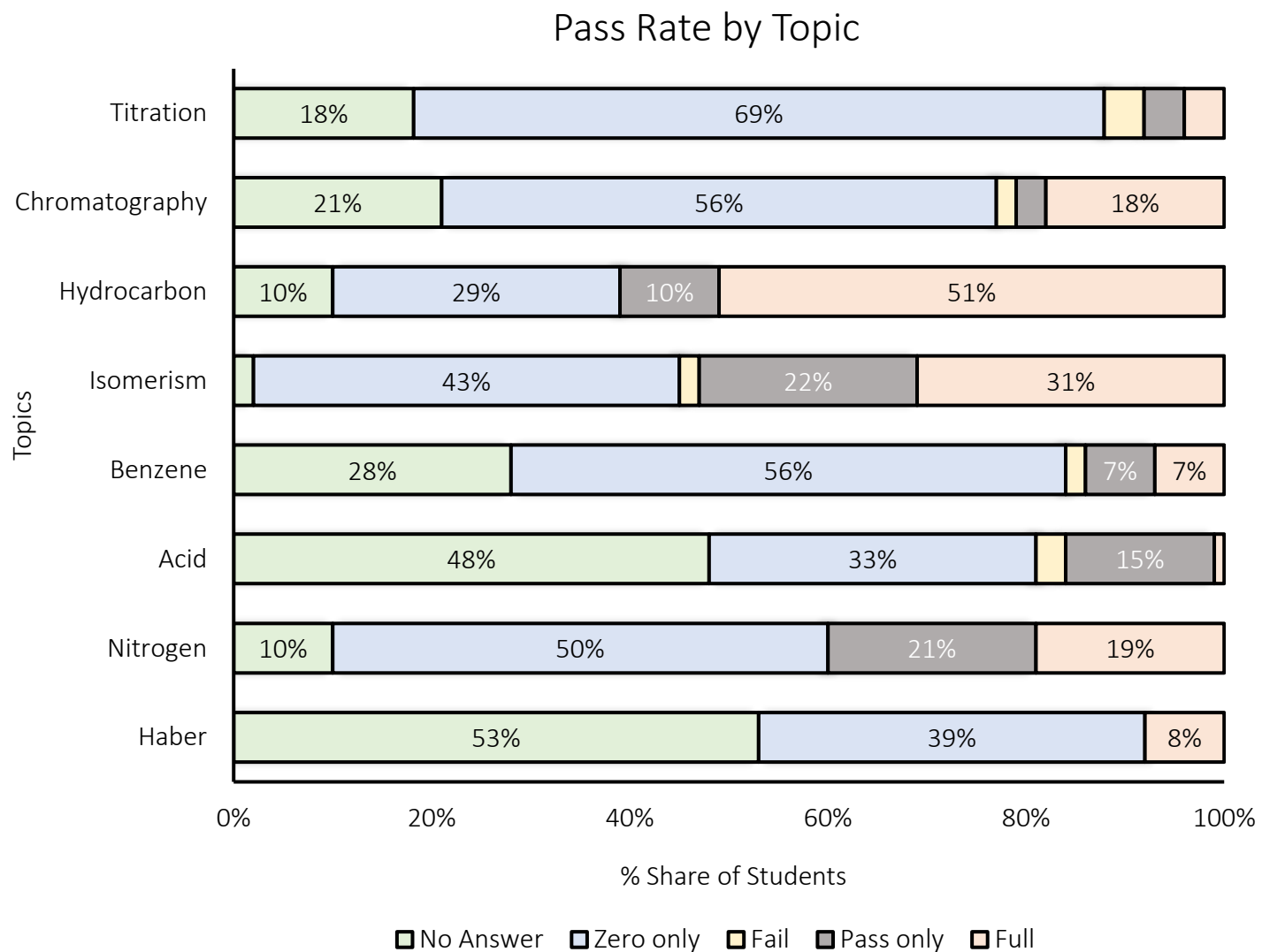
- Hydrocarbon was the highest performing topic for the top 5 performers as well as the bottom 5 performers
- Both top & bottom had similar performance patterns

Performance by Topic of Top 5 & Bottom 5 Students





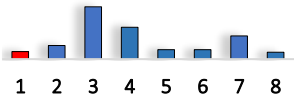
Pass Rate by Topic



- Titration questions had the largest share of zeros, while Hydrocarbon question had the largest share of full marks



Question 1 (Titration) Performance Analysis

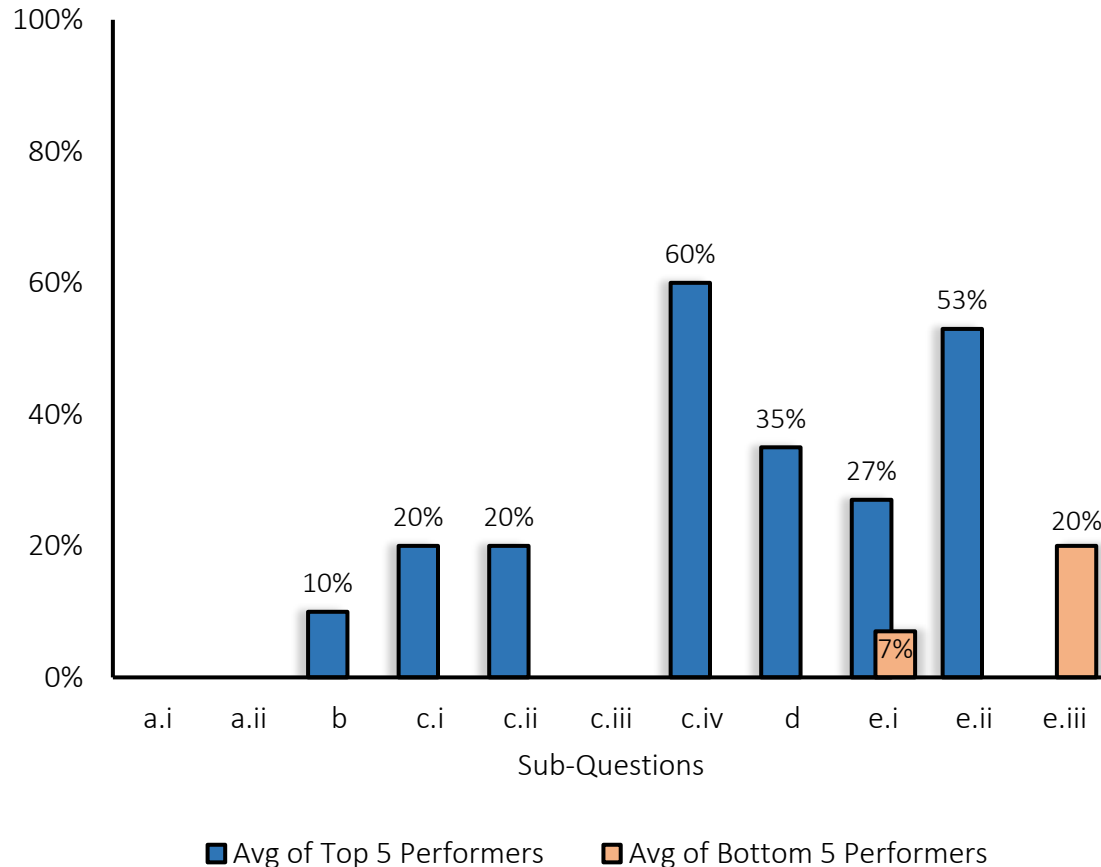


Question 1 - Titration



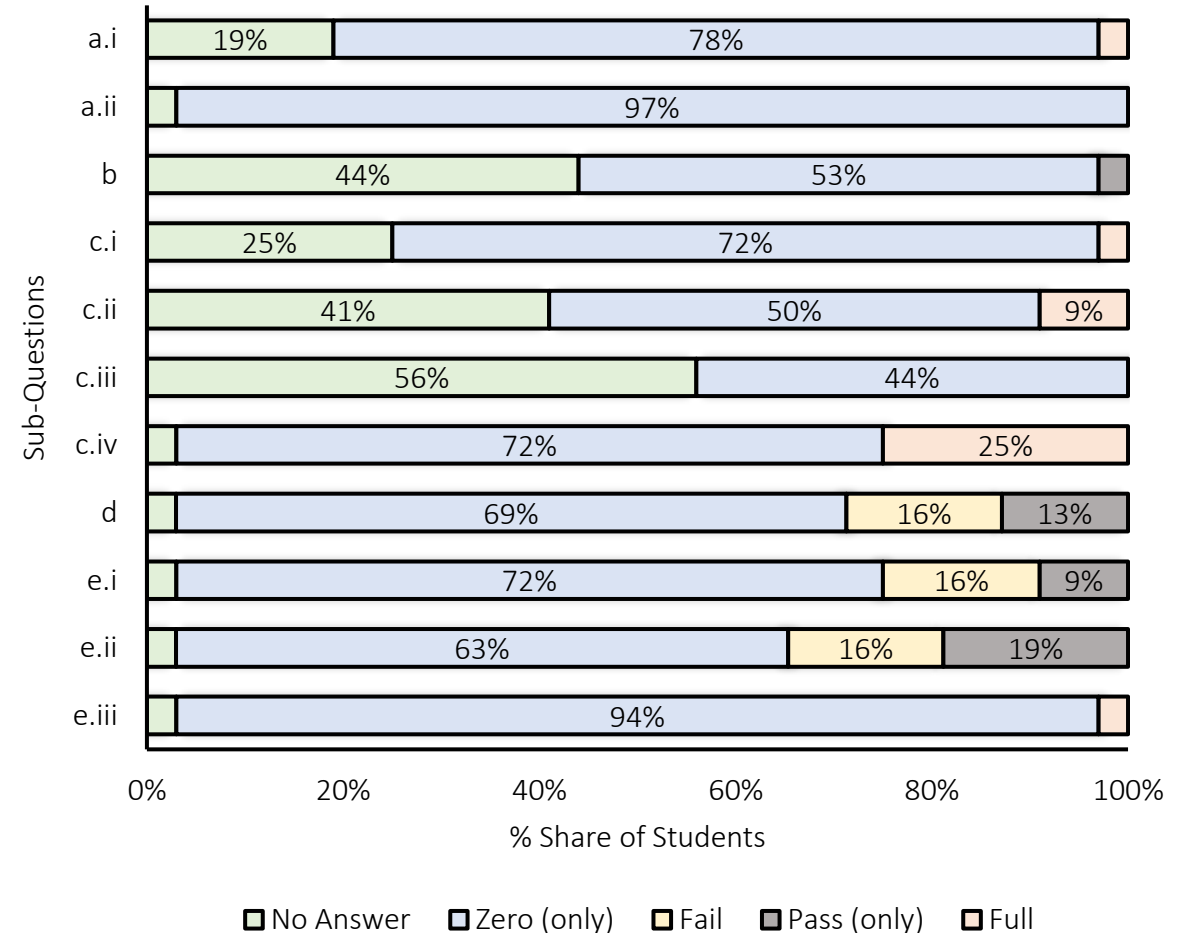
- Bottom 5 students averaged zero for parts a - d

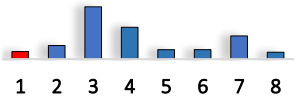
Question 1 (Titration) Top 5 & Bottom 5 Avg Scores



- Part a.ii receive the most zeros, while part c.iv received the most full-marks

Pass Rate for Question 1 (Titration)

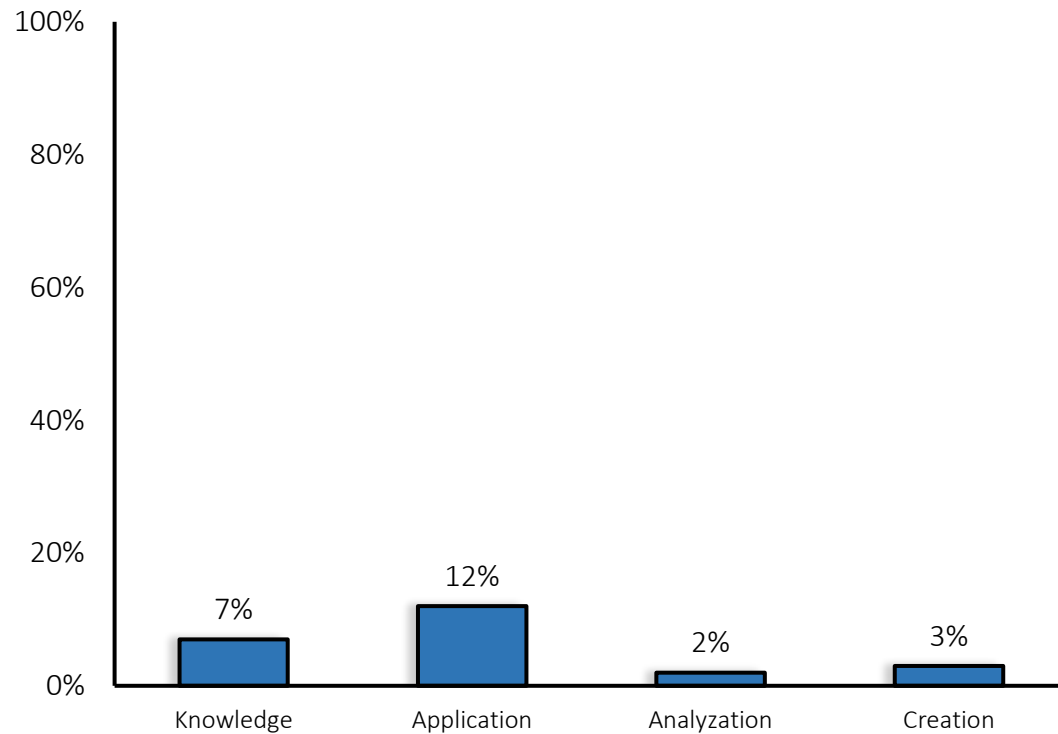




Question 1 - Titration

- Application had the highest average score – 12%

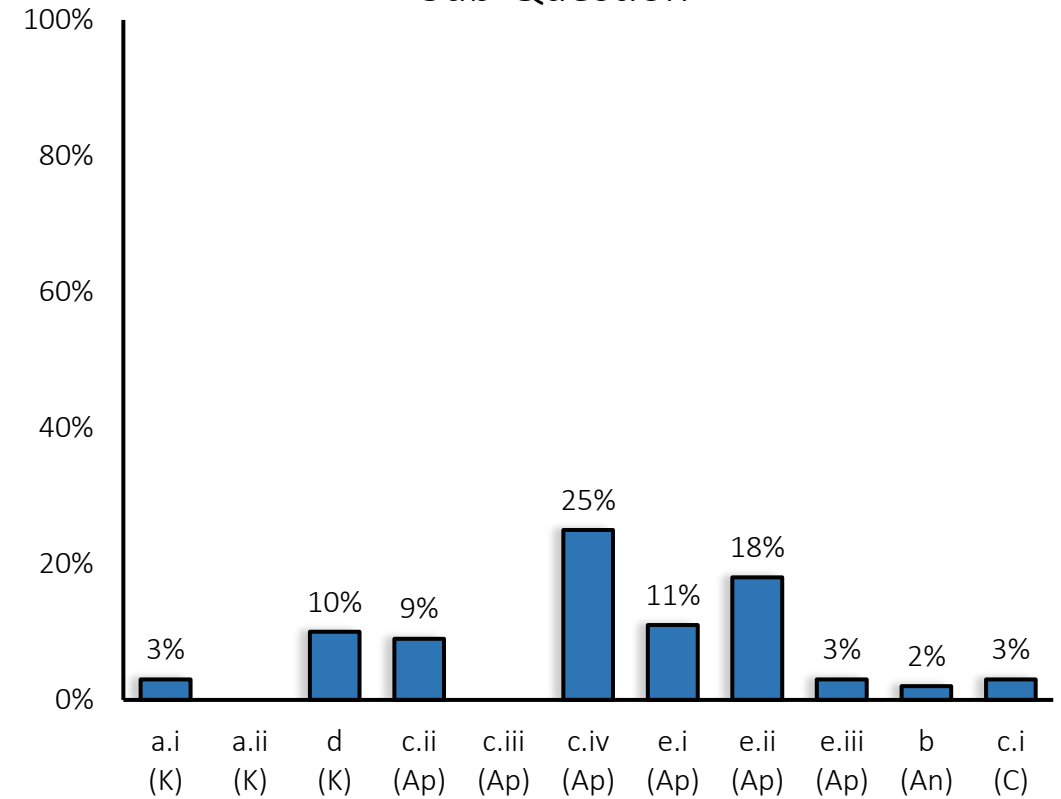
Question 1 (Titration) Average Score by Taxonomy



Question Taxonomy

- Performance of knowledge-based questions averaged 7%

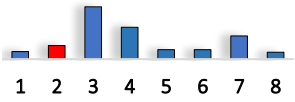
Question 1 (Titration) Avg Score by Sub-Question



Sub-Questions



Question 2 (Chromatography) Performance Analysis

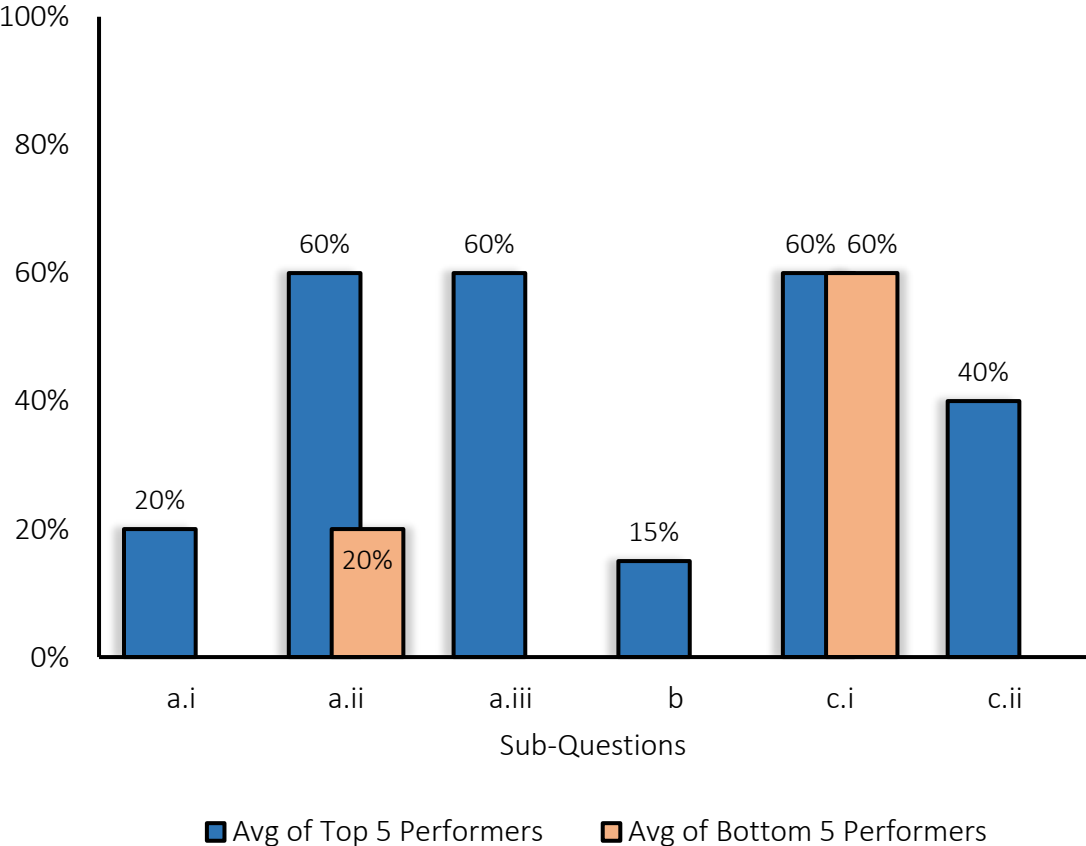


Question 2 - Chromatography



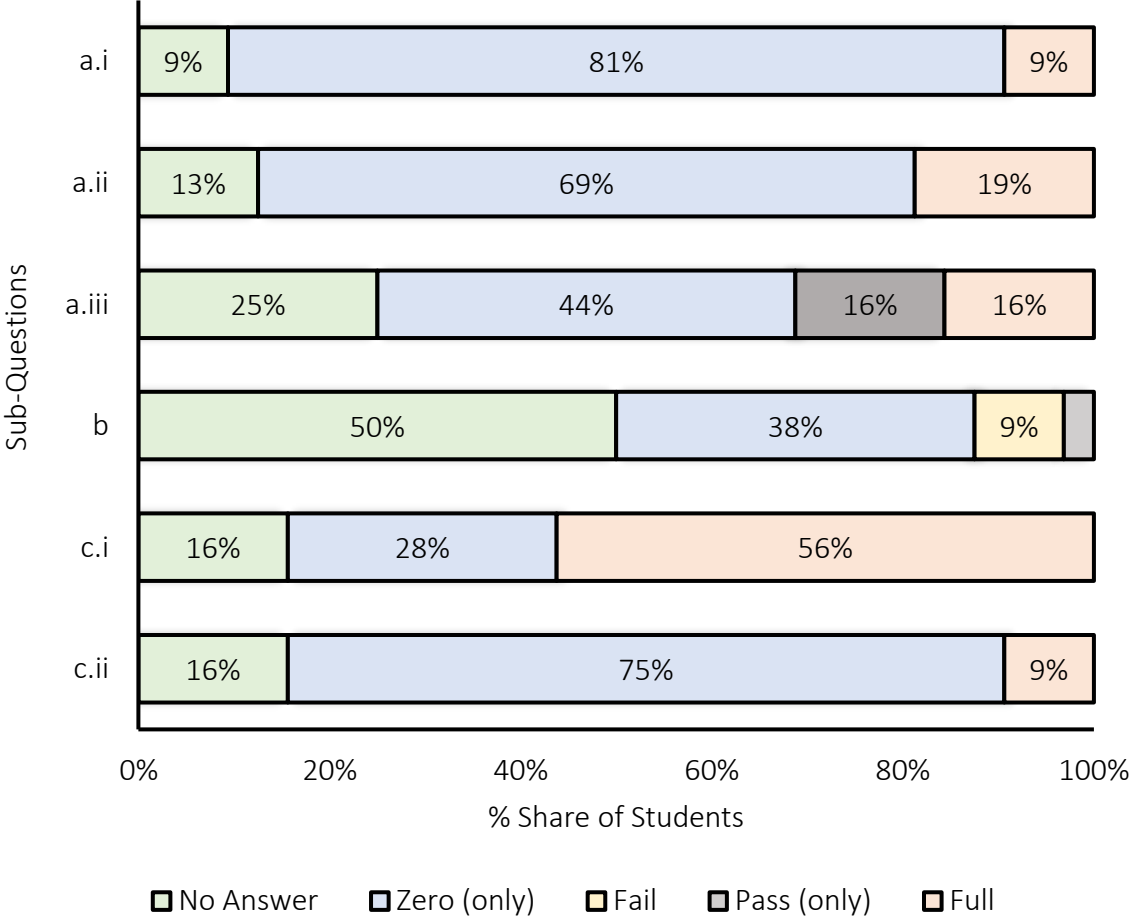
- Top & Bottom 5 averaged 60% on part c.i

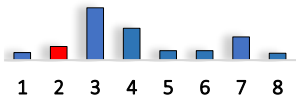
Question 2 (Chromatography) Top 5 & Bottom 5 Avg Scores



- Part a.i receive the most zeros, while part c.i received the most full-marks

Pass Rate for Question 2 (Chromatography)

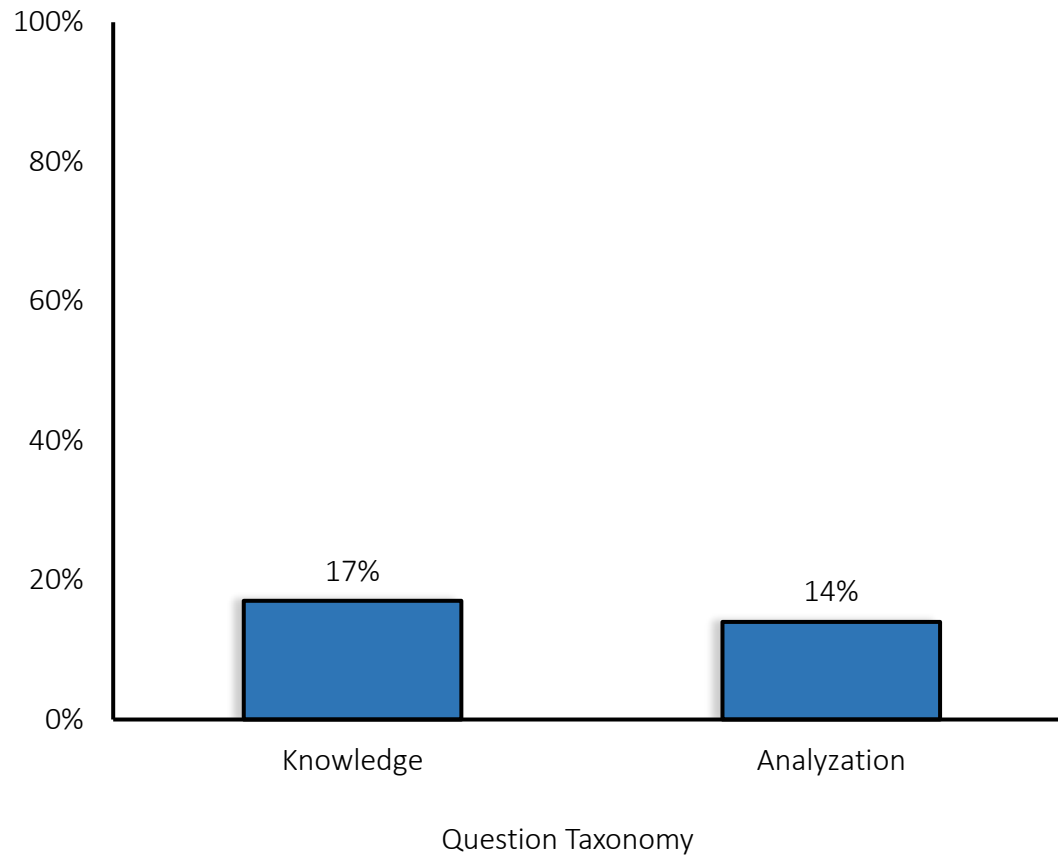




Question 2 - Chromatography

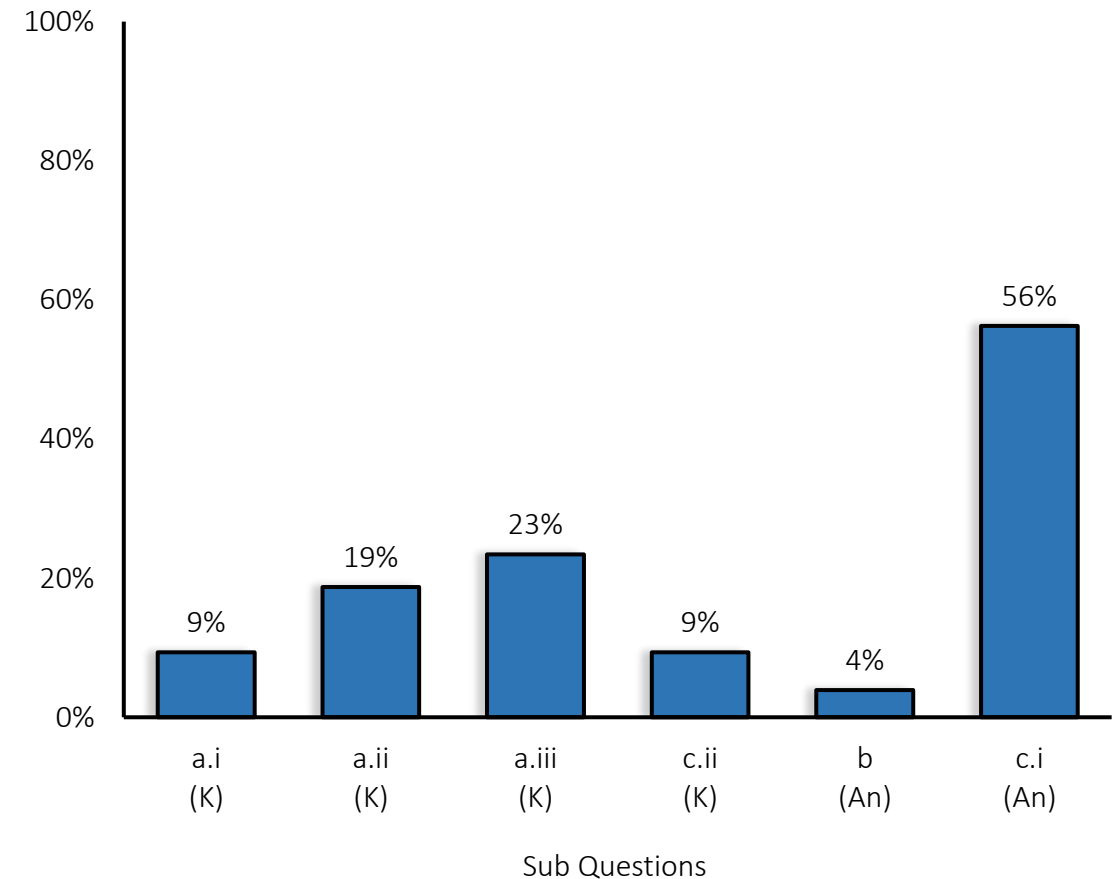
- Performance of Knowledge questions averaged 17%

Question 2 (Chromatography) Average Score by Taxonomy



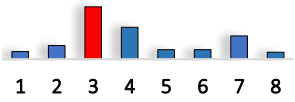
- Analyzation grades averaged below 14%

Question 2 (Chromatography) Avg Score by Sub-Question





Question 3 (Hydrocarbon) Performance Analysis

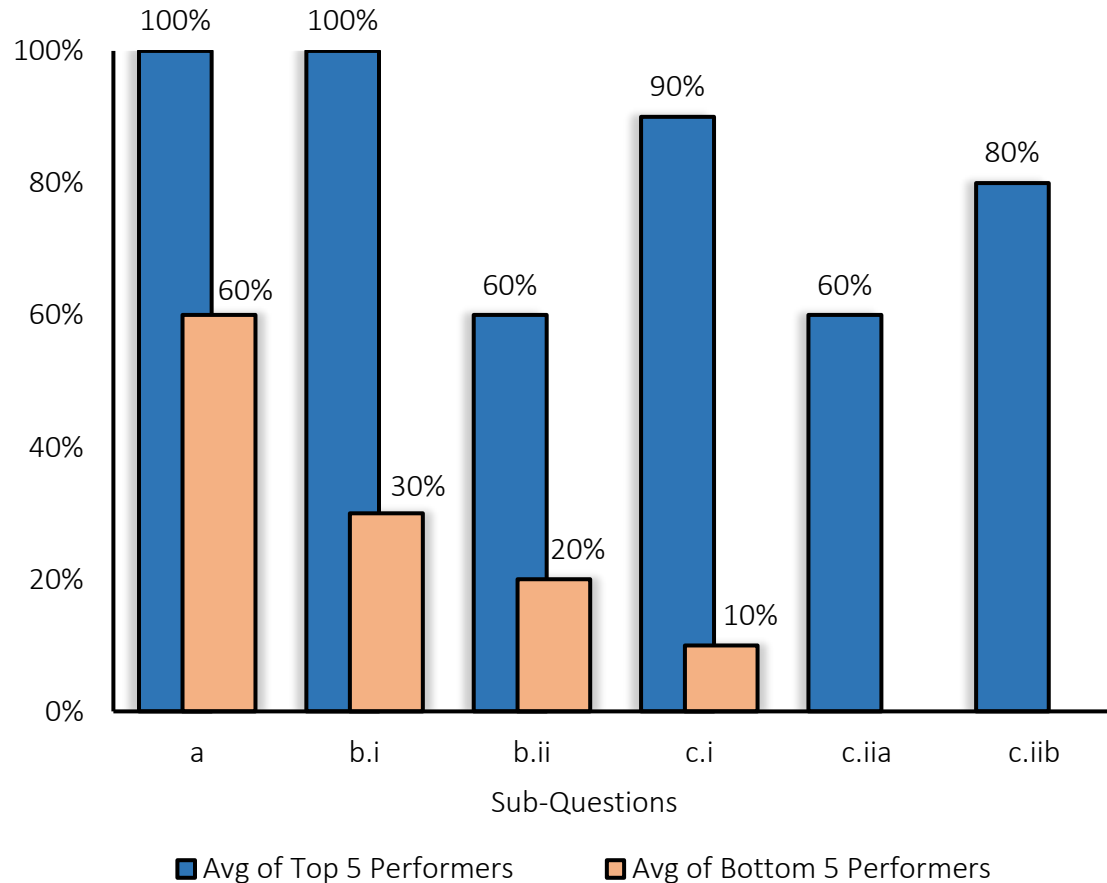


Question 3 - Hydrocarbon



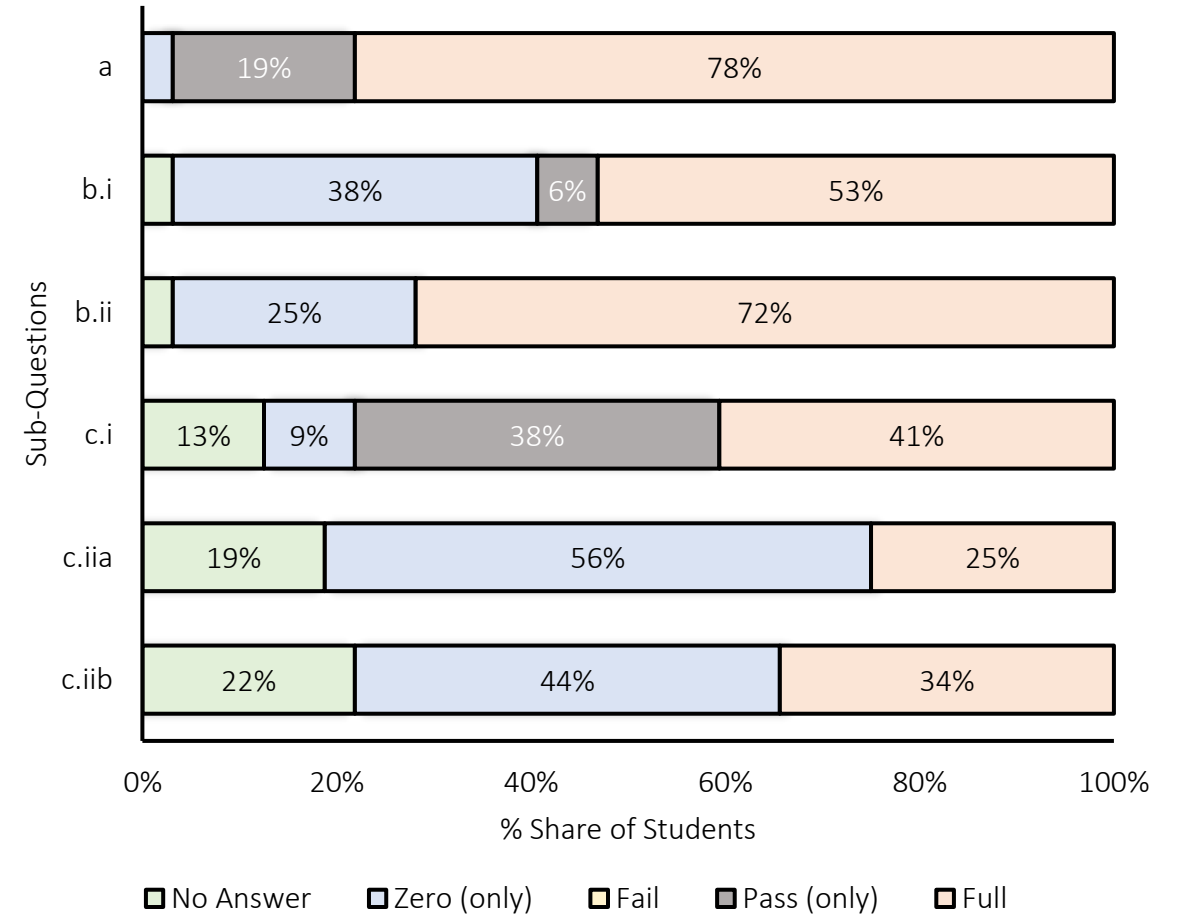
- Bottom 5 exhibited a downward trends from a - c

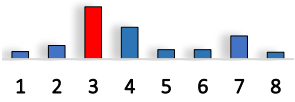
Question 3 (Hydrocarbon) Top 5 & Bottom 5
Avg Scores



- Part c.ii.a received most zeros, while part a received highest share of full marks

Pass Rate for Question 3 (Hydrocarbon)



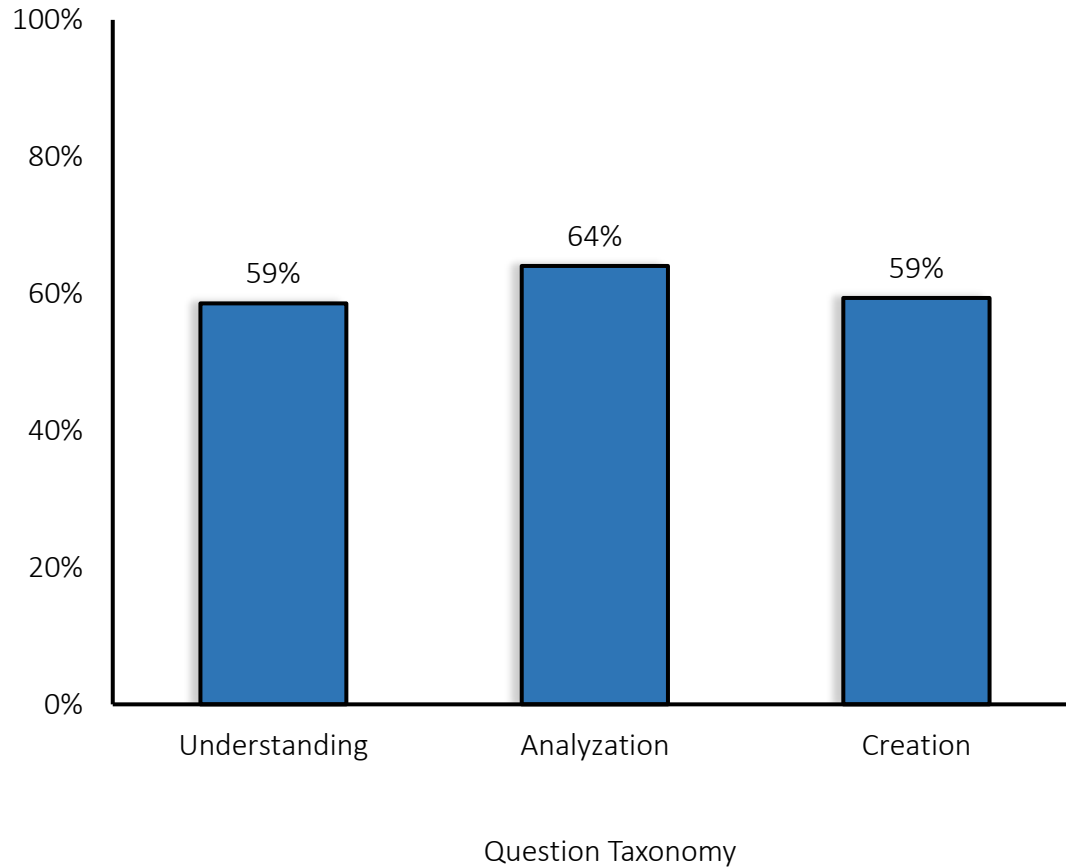


Question 3 - Hydrocarbon



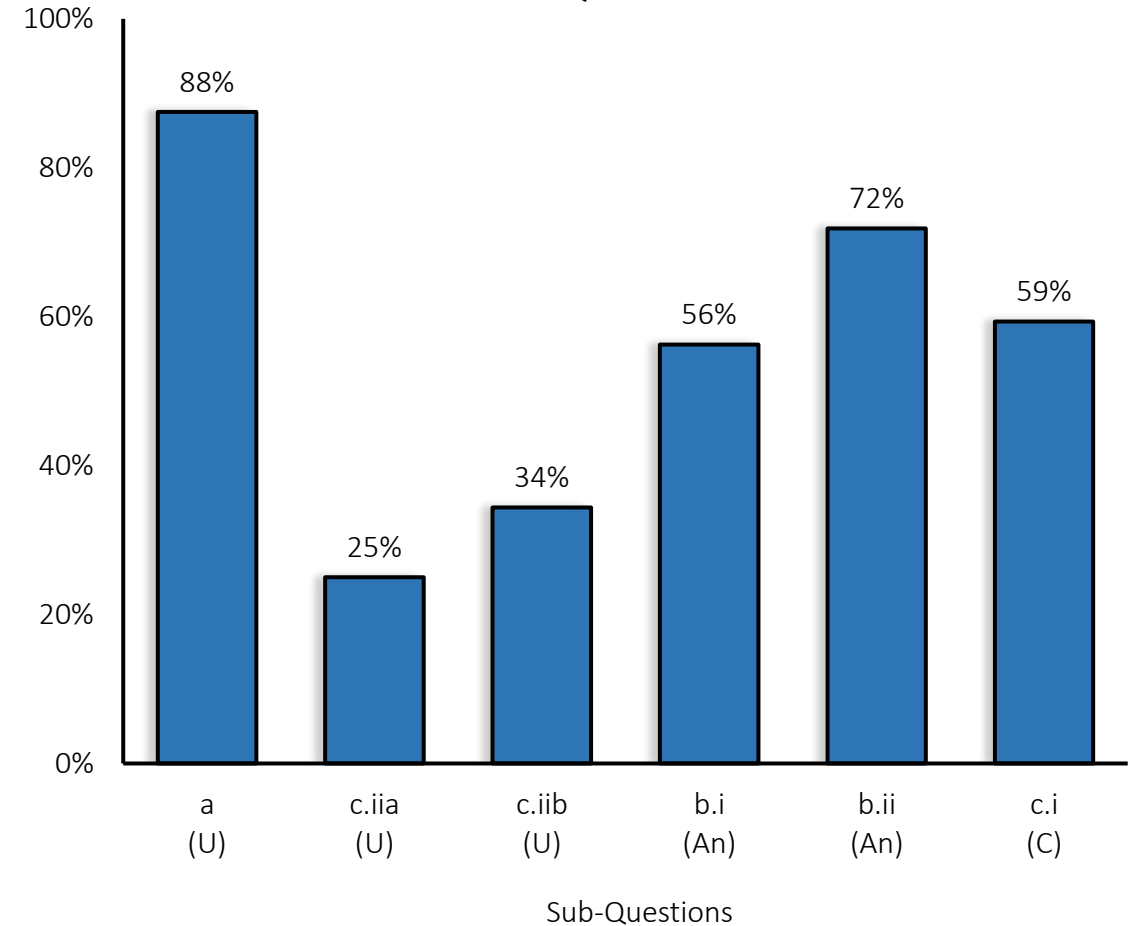
- Analyzation had the highest average score – 32%

Question 3 (Hydrocarbon) Avg Score by Taxonomy



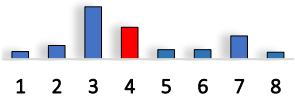
- Understand grades ranged from 25% to 88%

Question 3 (Hydrocarbon) Avg Score by Sub-Question





Question 4 (Isomerism) Performance Analysis

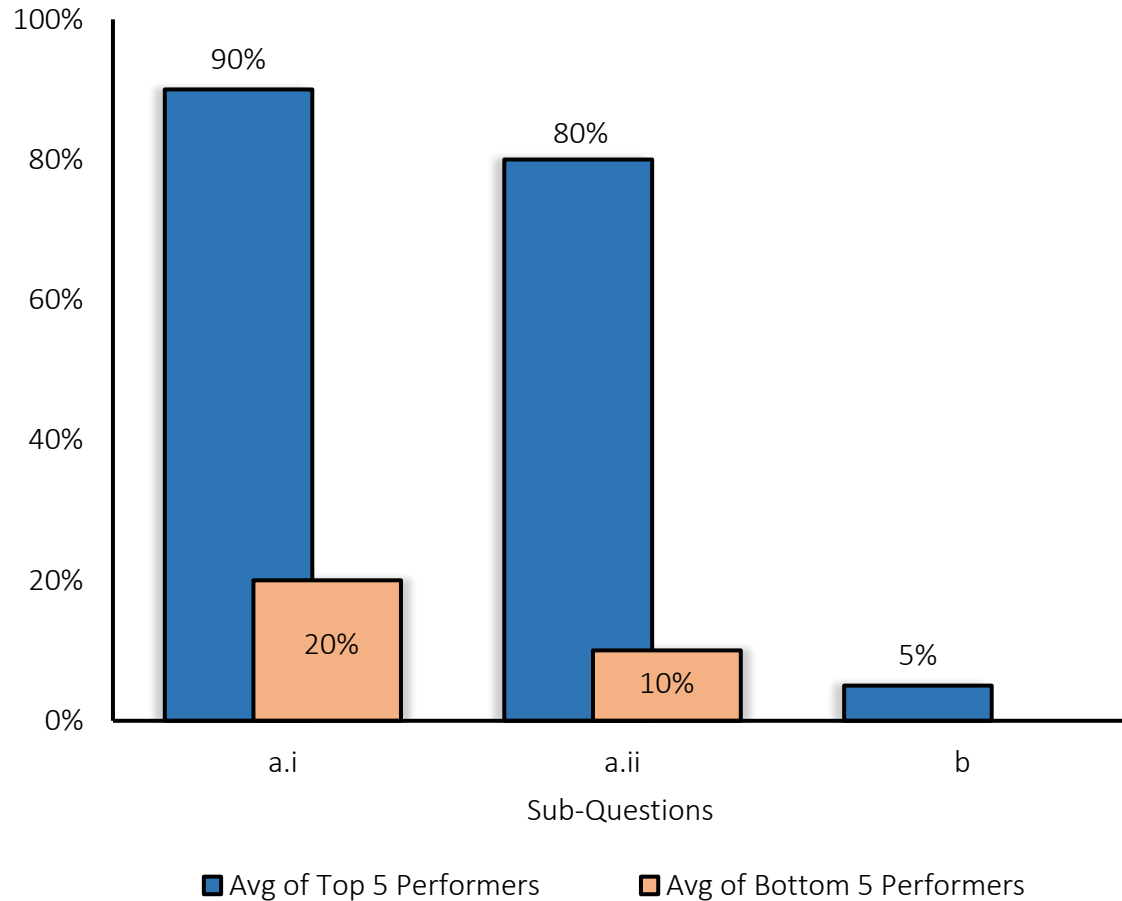


Question 4 - Isomerism



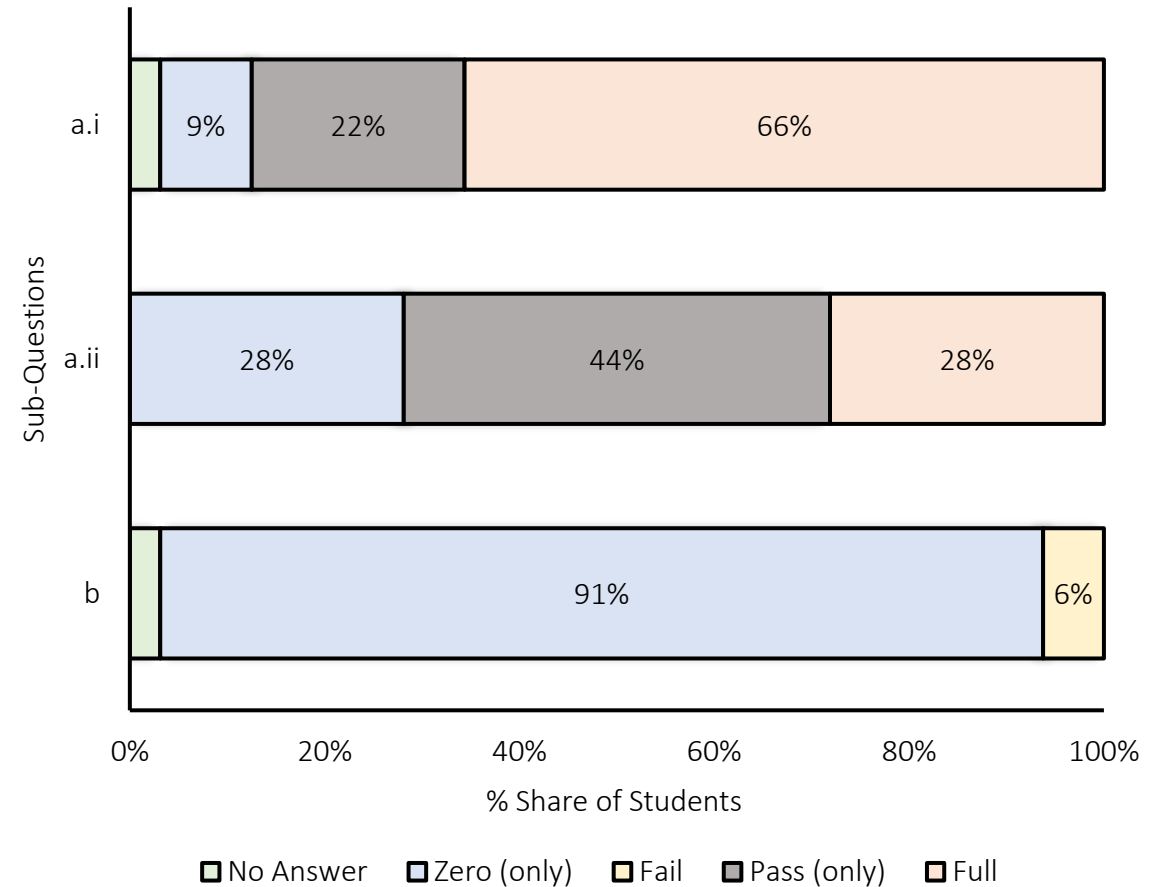
- Bottom 5 performers averaged 7%, while top 5 averaged of 30%

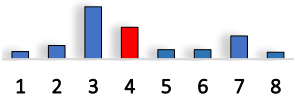
Question 4 (Isomerism) Top 5 & Bottom 5 Avg Scores



- Part **b** received the highest share of zeros, while part **a.i** received most full-marks

Pass Rate for Question 4 (Isomerism)



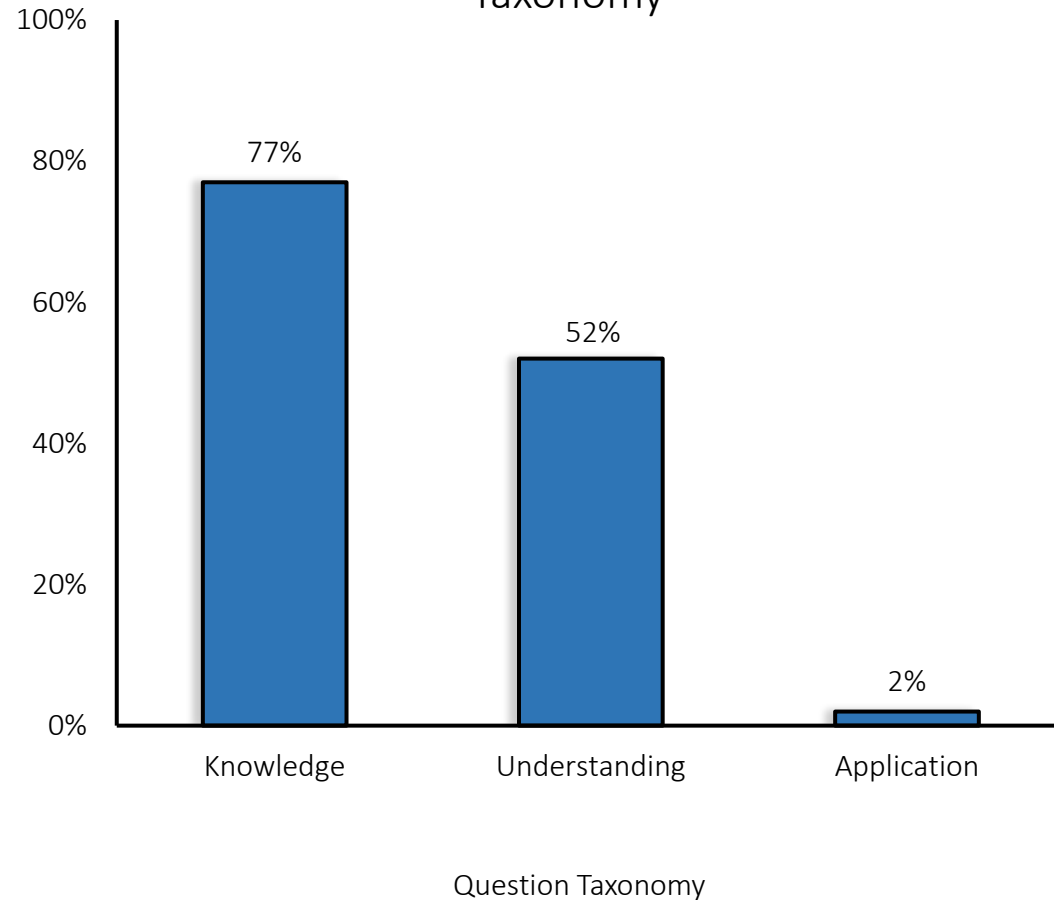


Question 4 - Isomerism



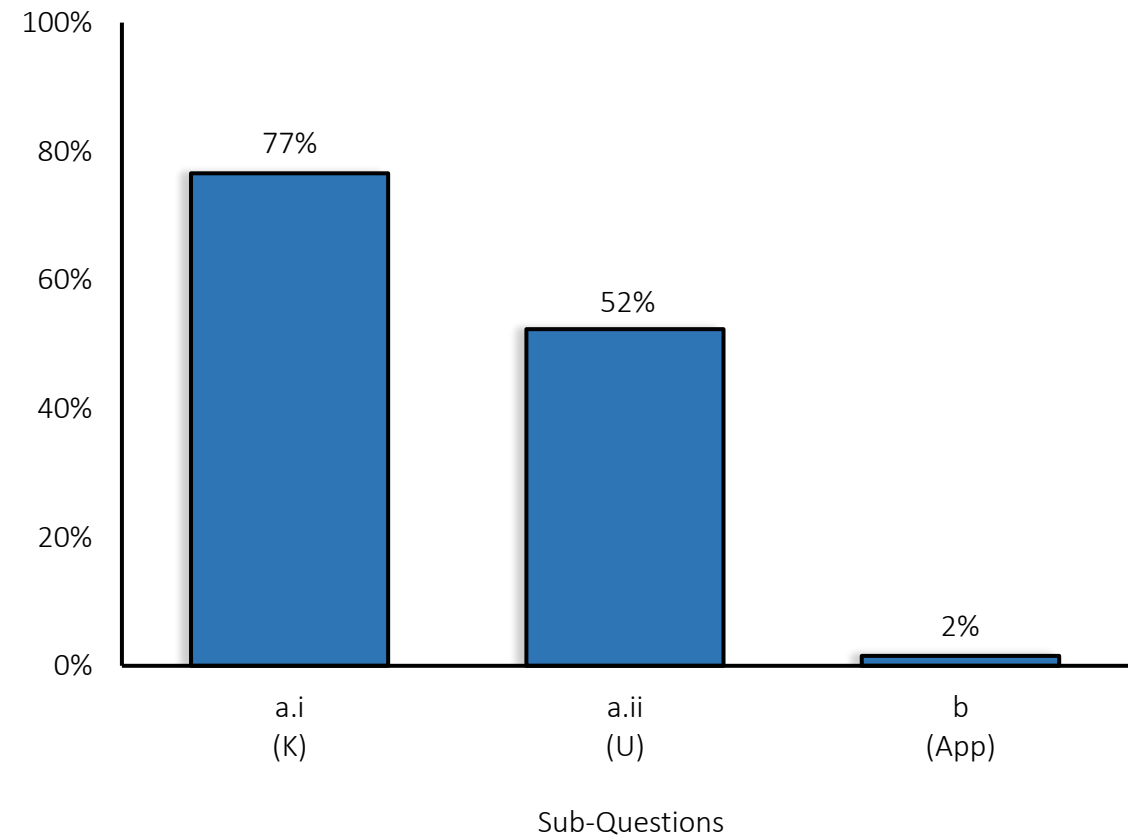
- Knowledge had the highest average score – 77%

Question 4 (Isomerism) Avg Score by Taxonomy



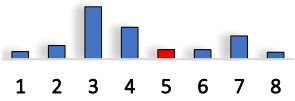
- Understanding averaged 52%

Question 4 (Isomerism) Avg Score by Sub-Question





Question 5 (Benzene) Performance Analysis

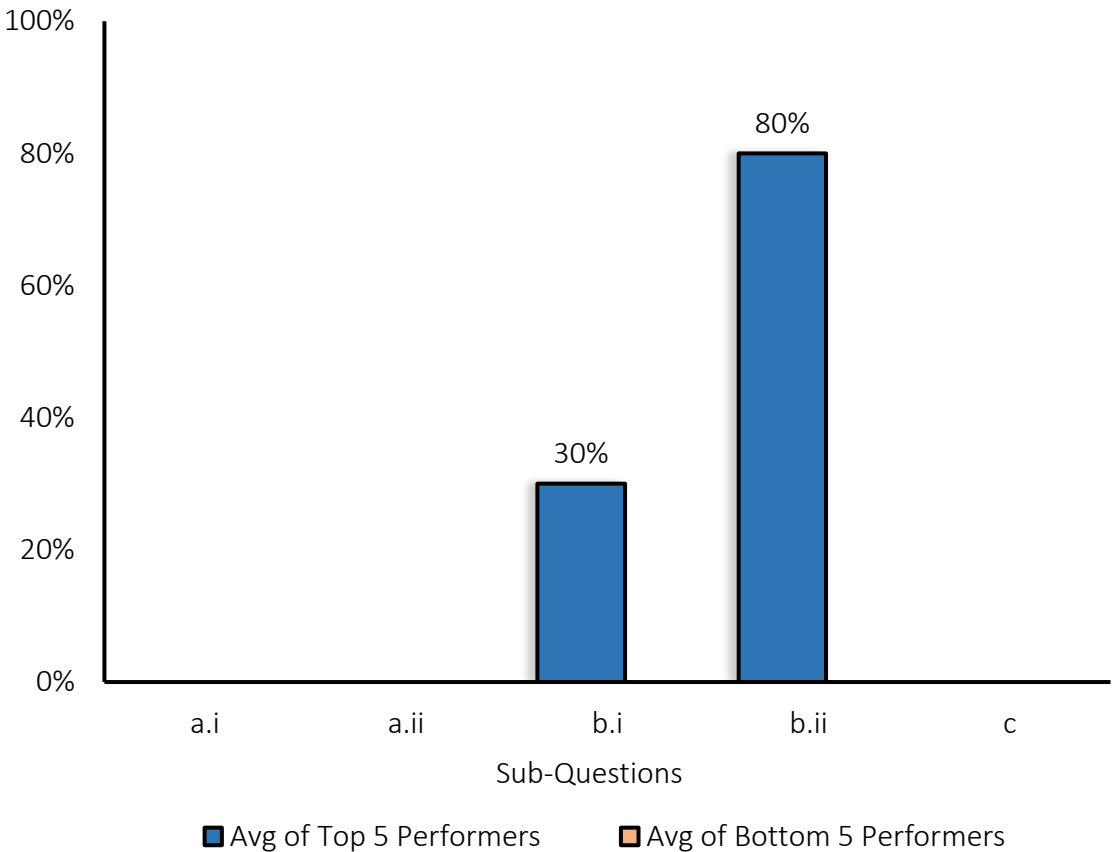


Question 5 - Benzene



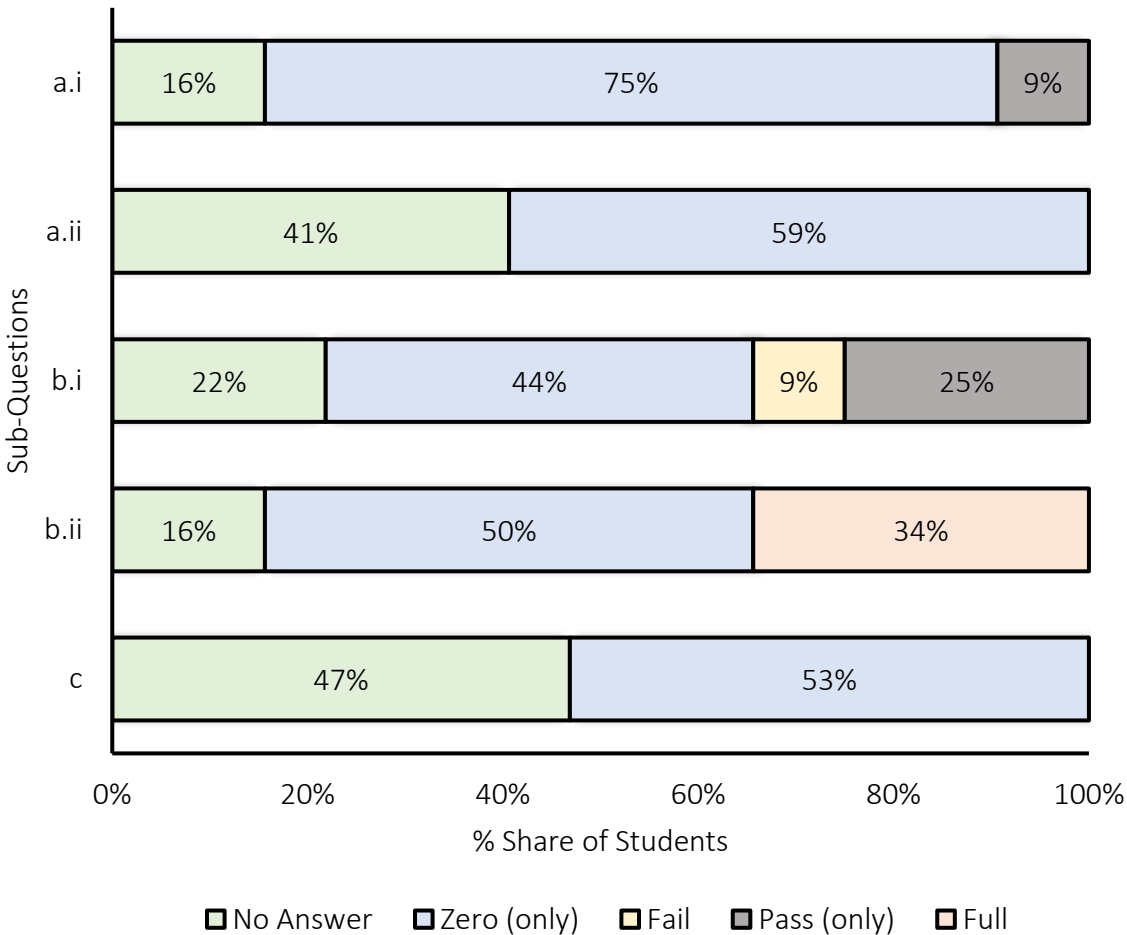
- Bottom 5 performers received an average score of zero for Benzene

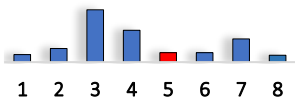
Question 5 (Benzene) Top 5 & Bottom 5
Avg Scores



- Part a.ii and c received all zeros and unattempted

Pass Rate for Question 5 (Benzene)

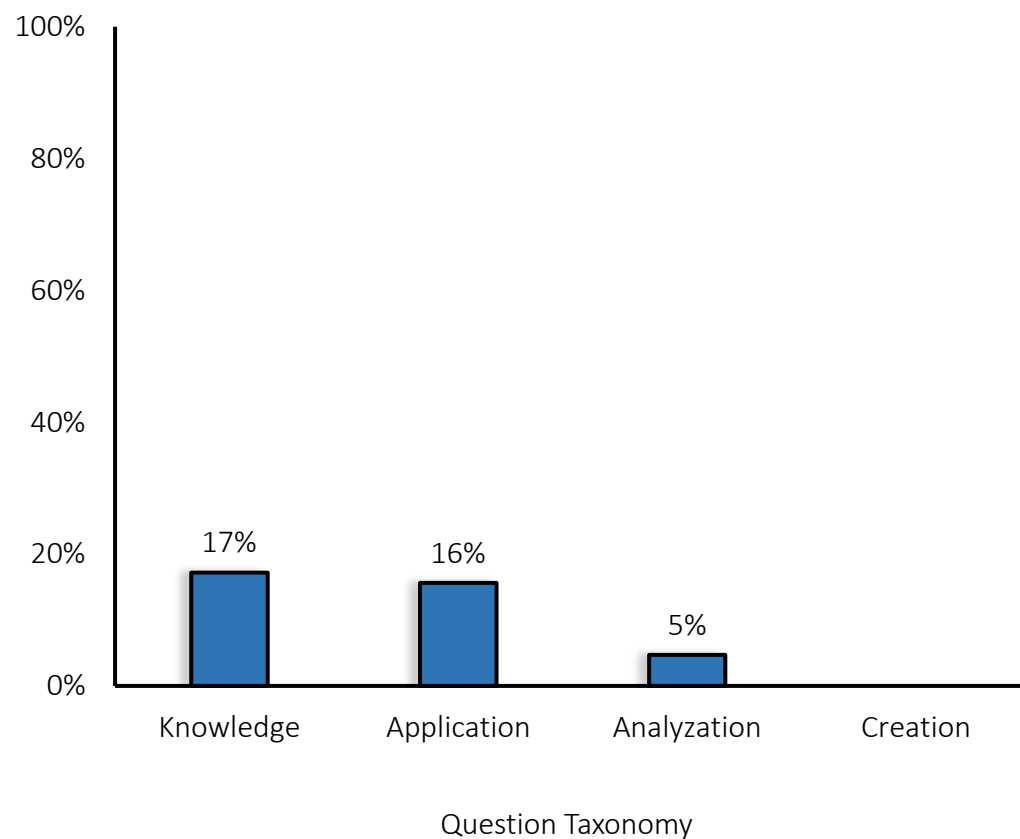




Question 5 - Benzene

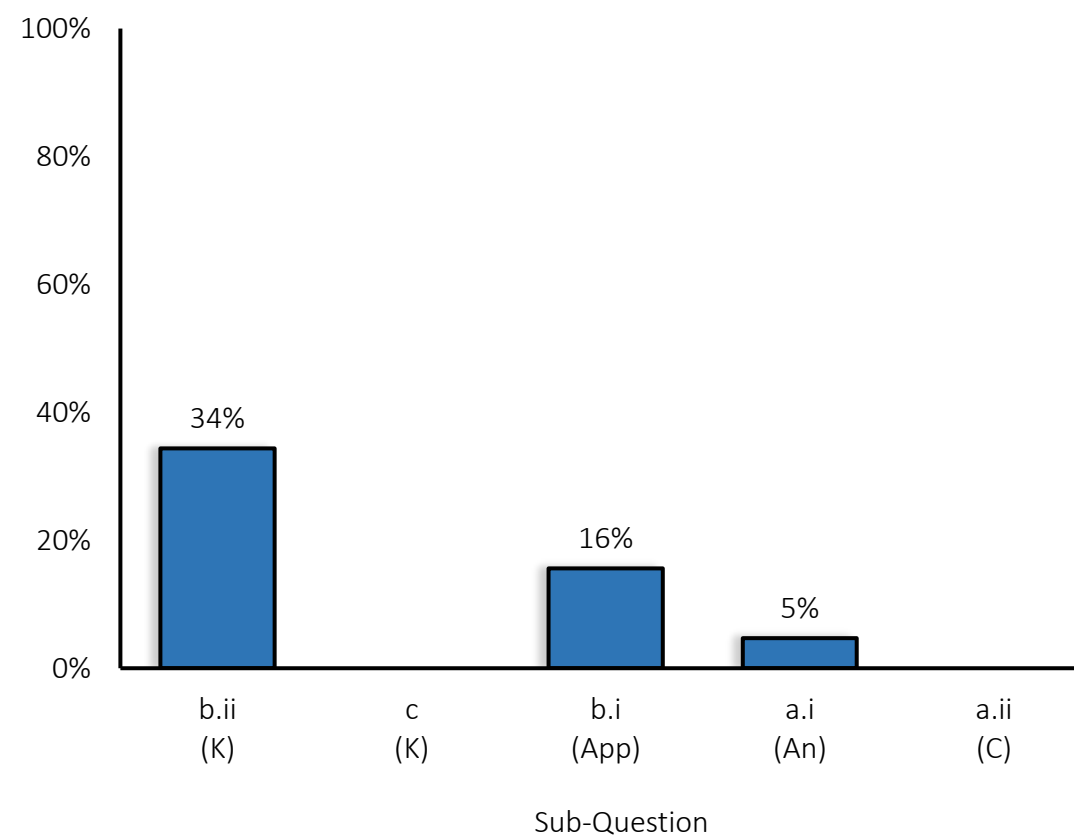
- Knowledge had the highest average score – 17%

Question 5 (Benzene) Avg Score by Taxonomy



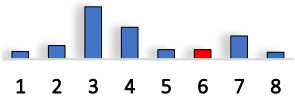
- Creation averaged the lowest grades – 0%

Question 5 (Benzene) Avg Score by Sub-Question





Question 6 (Acid) Performance Analysis

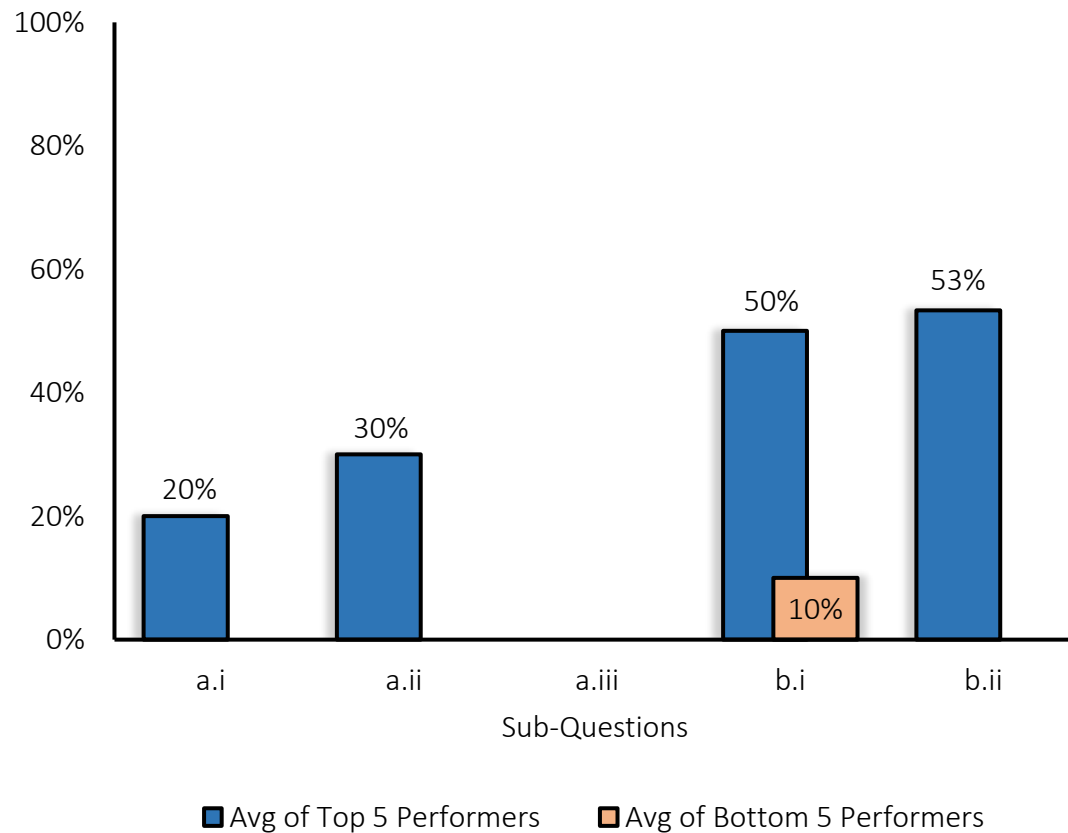


Question 6 - Acid



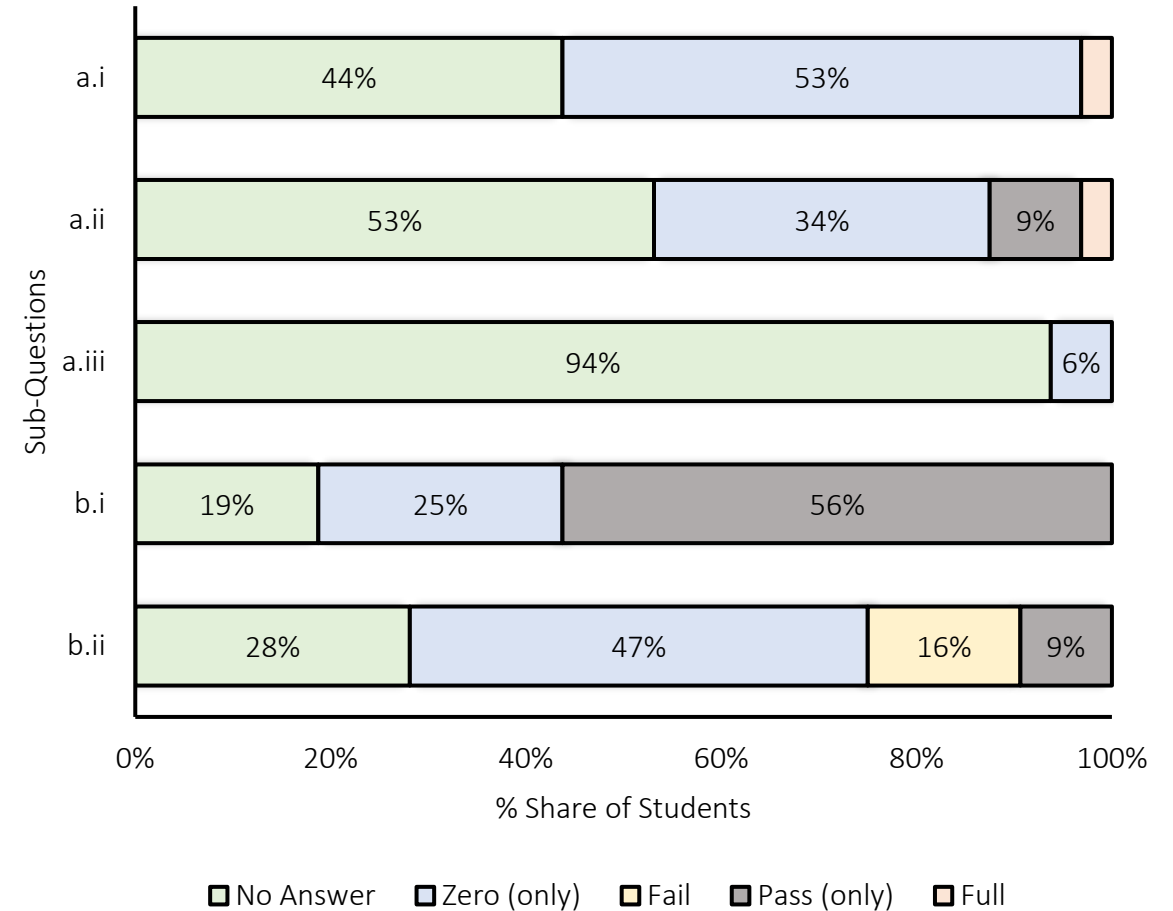
- Bottom 5 performers averaged 10% for part **b.i** but zero for the remaining parts

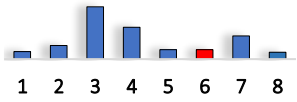
Question 6 (Acid) Top 5 & Bottom 5 Avg Scores



- Part **a.iii** received 94% non-responses

Pass Rate for Question 6 (Acid)



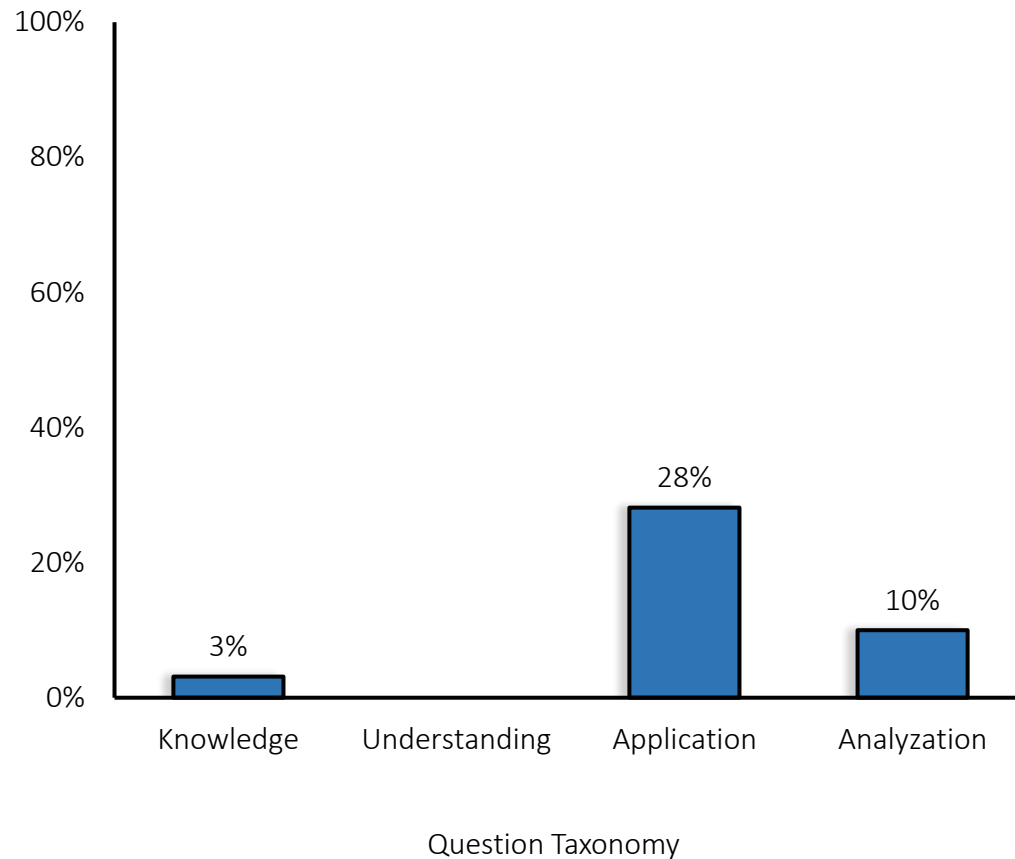


Question 6 - Acid



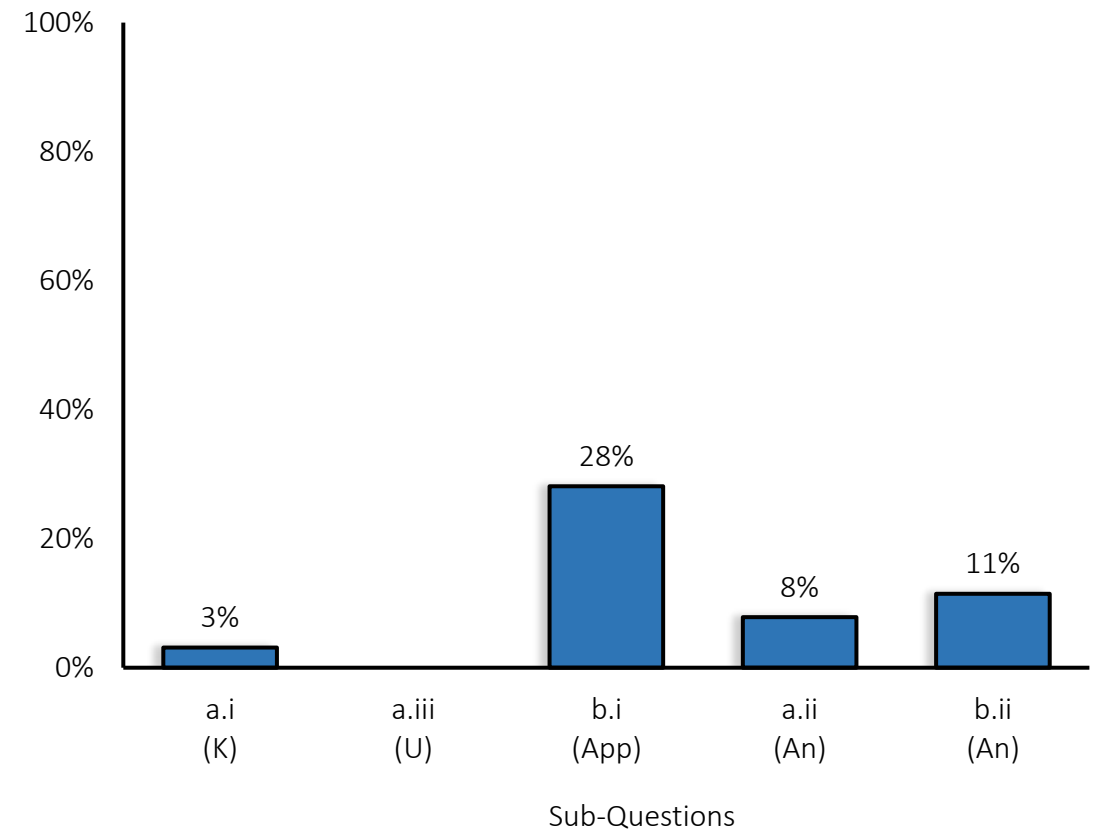
- Application had the highest average score 28%

Question 6 (Acid) Avg Score by Taxonomy



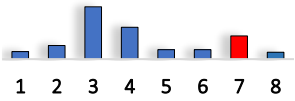
- Understand had an average score of 0%

Question 6 (Acid) Avg Score by Sub-Question





Question 7 (Nitrogen) Performance Analysis

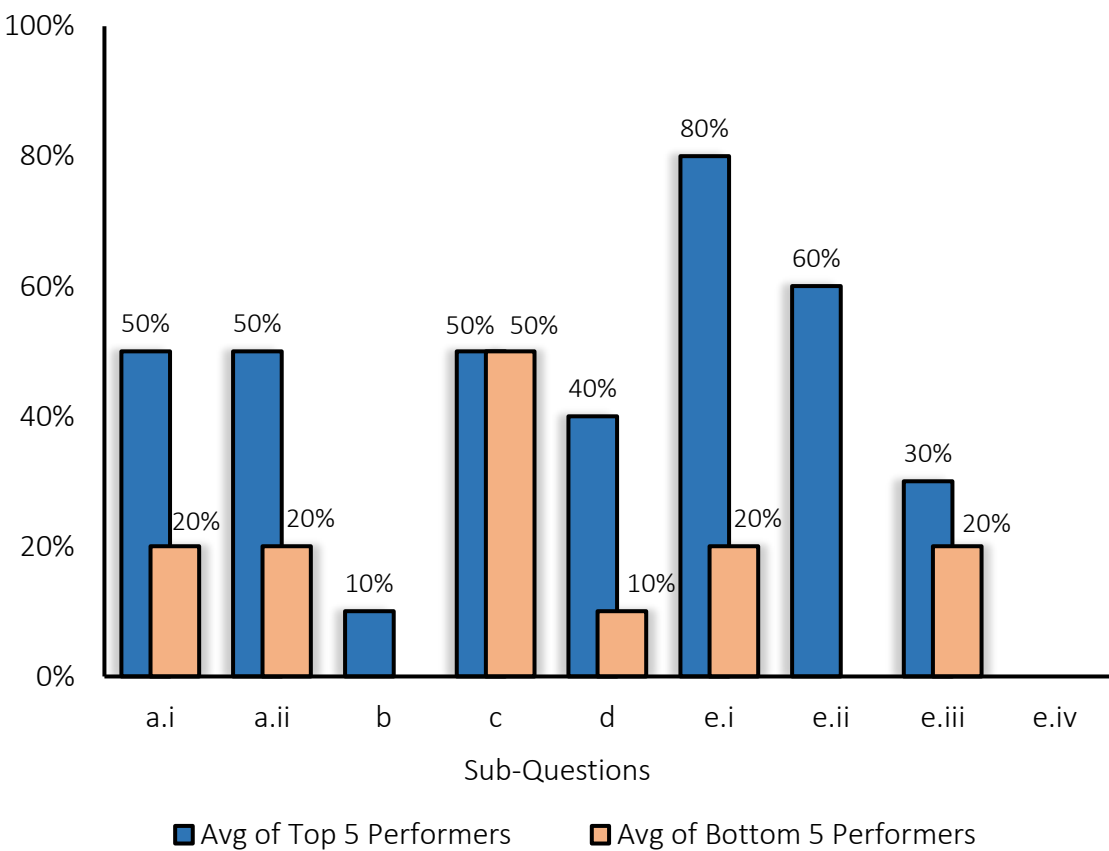


Question 7 - Nitrogen



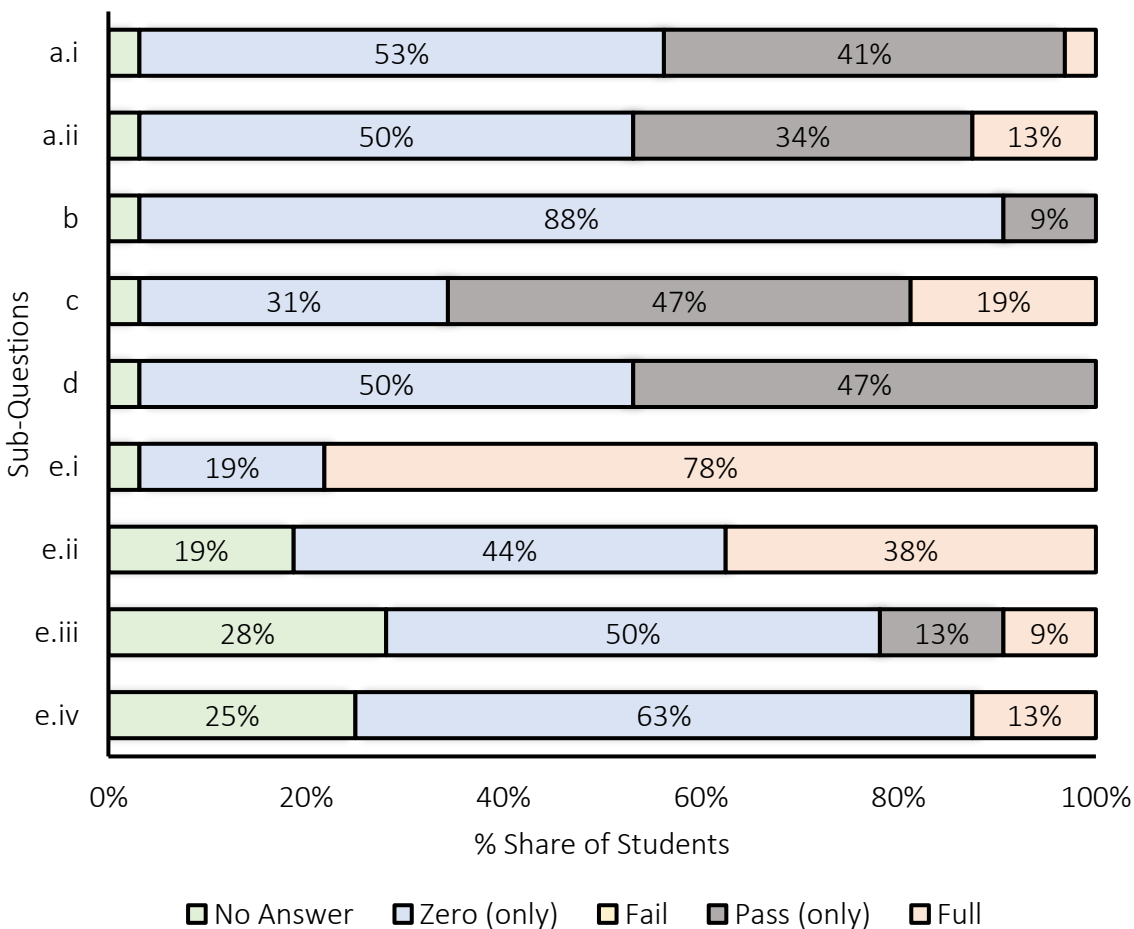
- Top & Bottom 5 performers received average score of zero in part **e.iv**

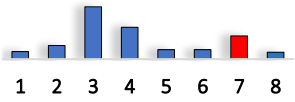
Question 7 (Nitrogen) Top 5 & Bottom 5 Avg Scores



- Part **c** received a high rate of zeros at 88%

Pass Rate for Question 7 (Nitrogen)



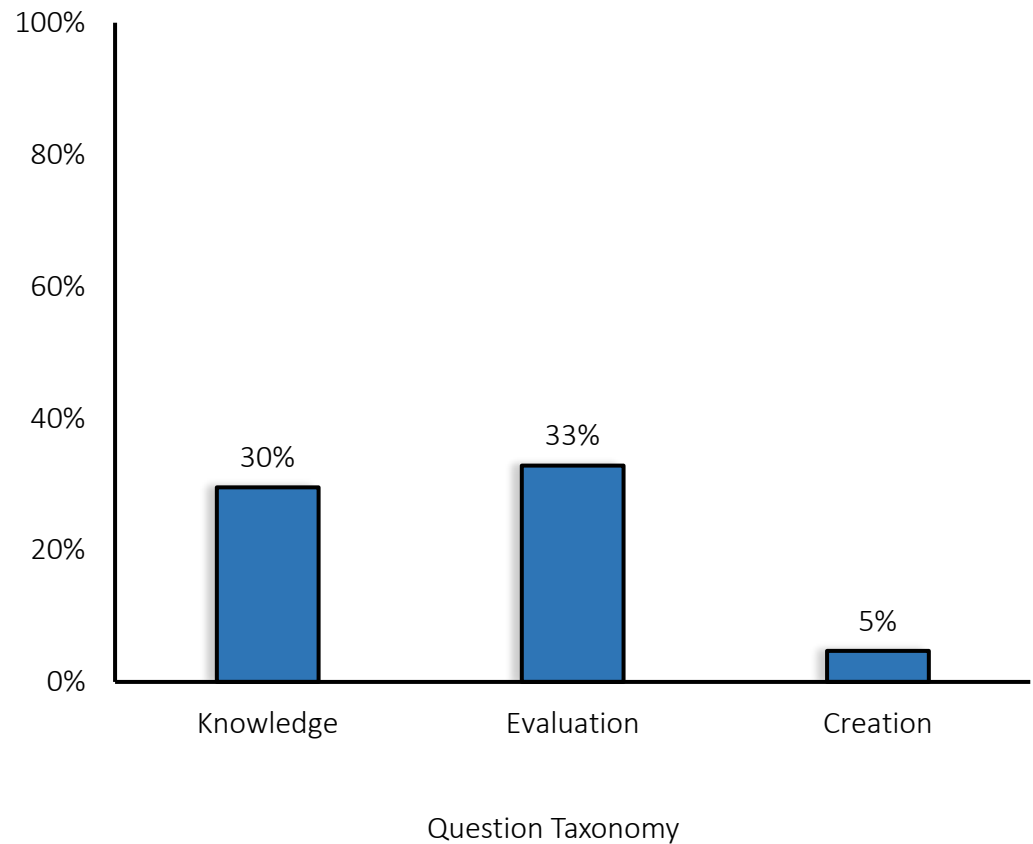


Question 7 - Nitrogen



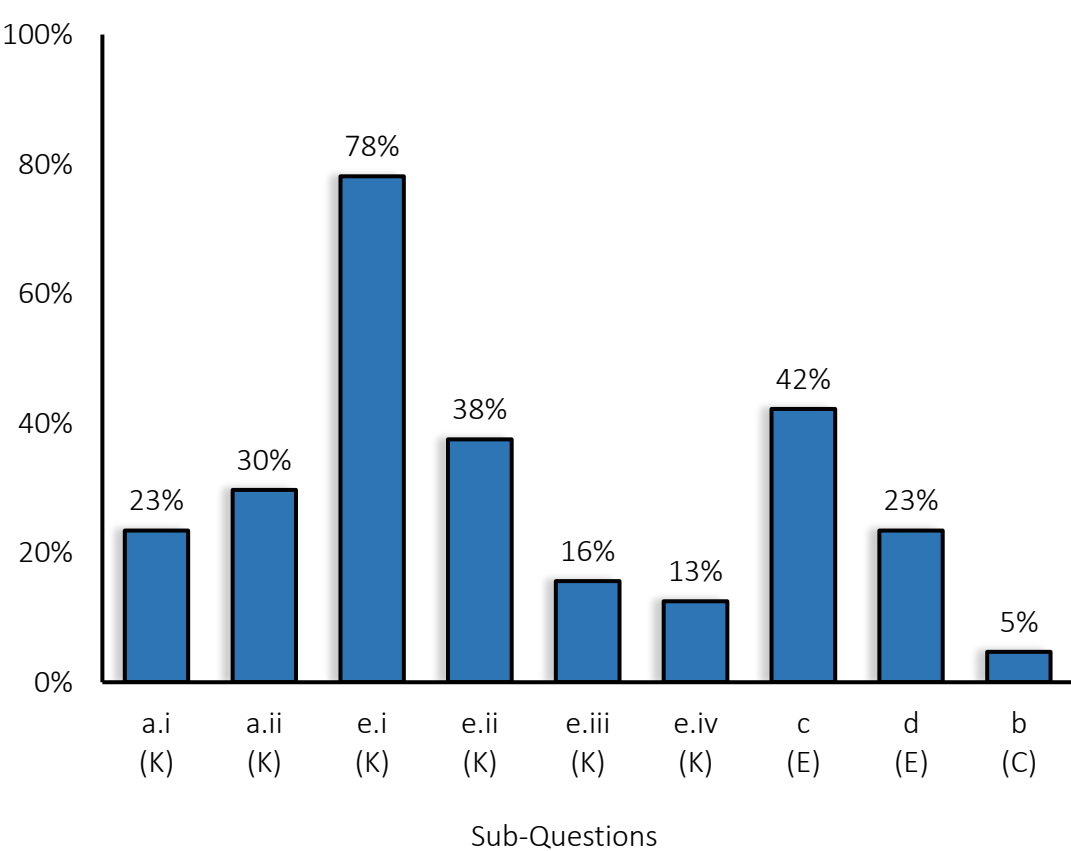
- Evaluation had the highest average score – 33%

Question 7 (Nitrogen) Avg Score by Taxonomy



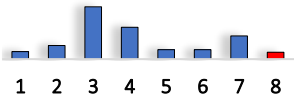
- Creation averaged the lowest grades – 5%

Question 7 (Nitrogen) Avg Score by Sub-Question





Question 8 (Haber Process) Performance

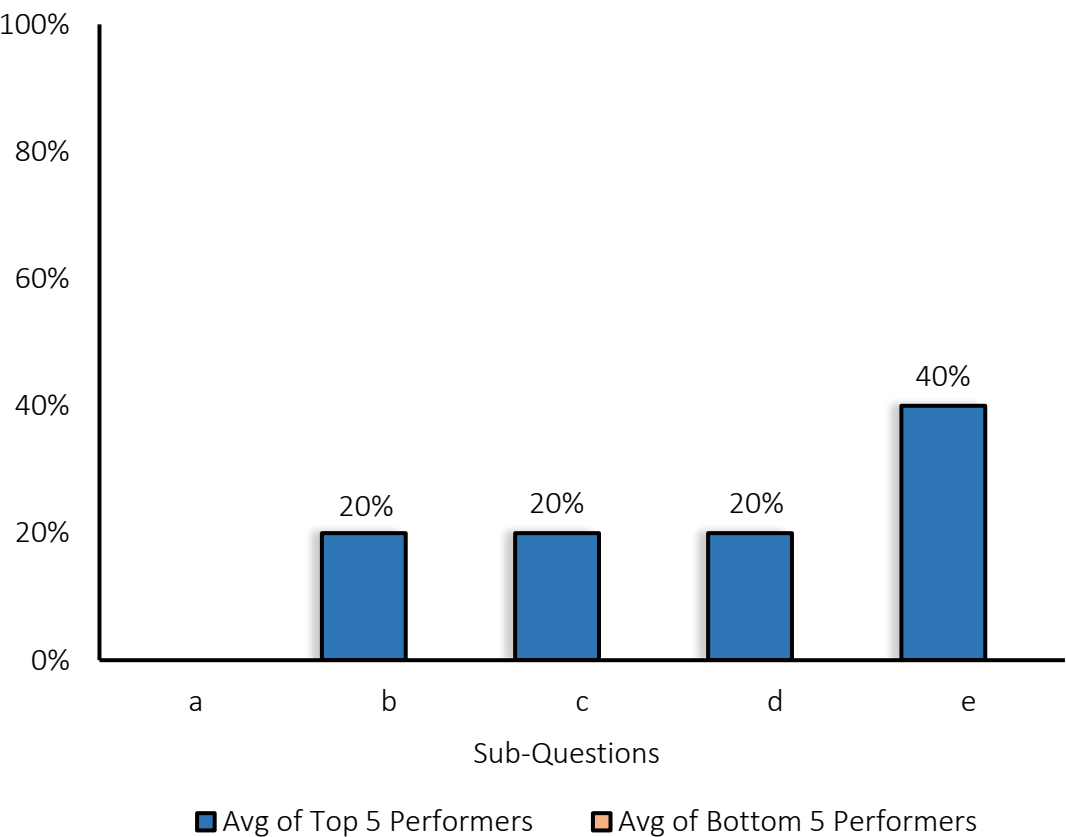


Question 8 - Haber



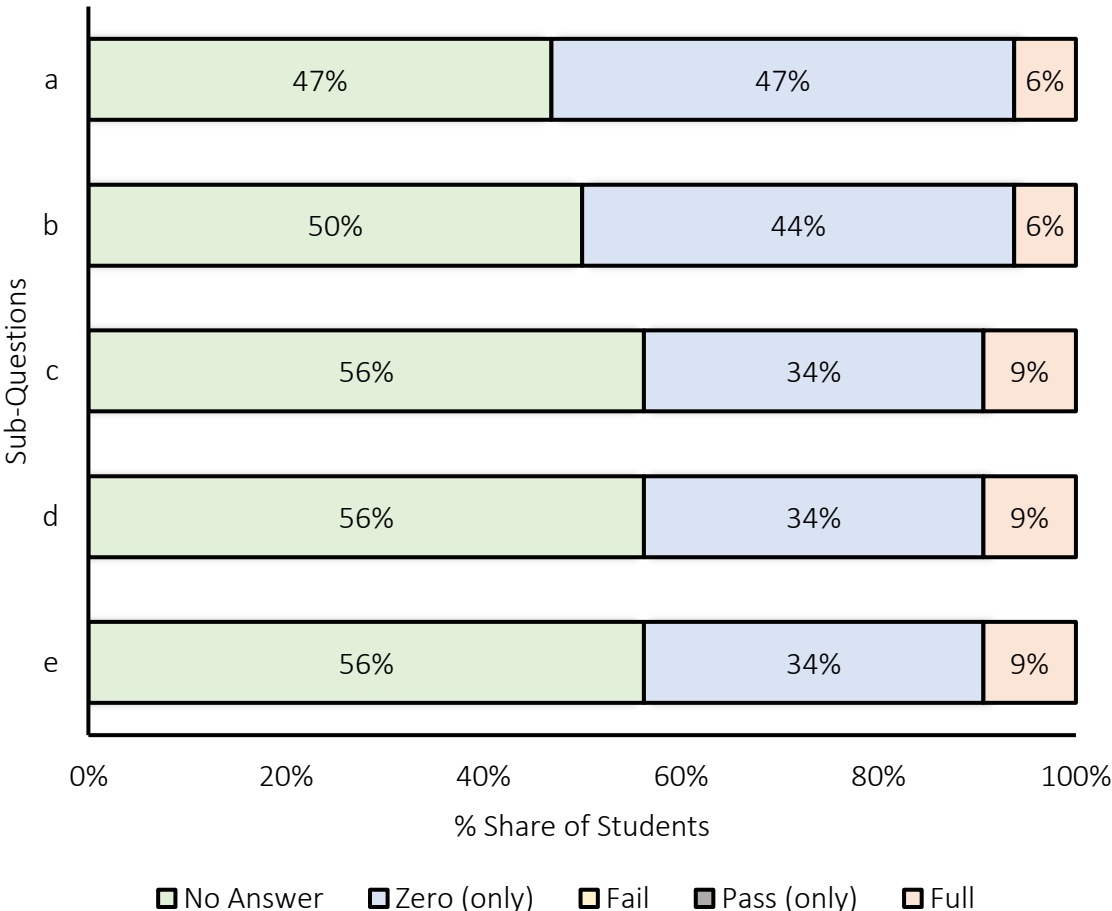
- Top & Bottom 5 performers averaged 0% for part a

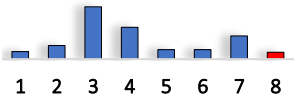
Question 8 (Haber) Top 5 & Bottom 5 Avg Scores



- More than 50% of class did not answer Part c - e

Pass Rate for Question 8 (Haber)



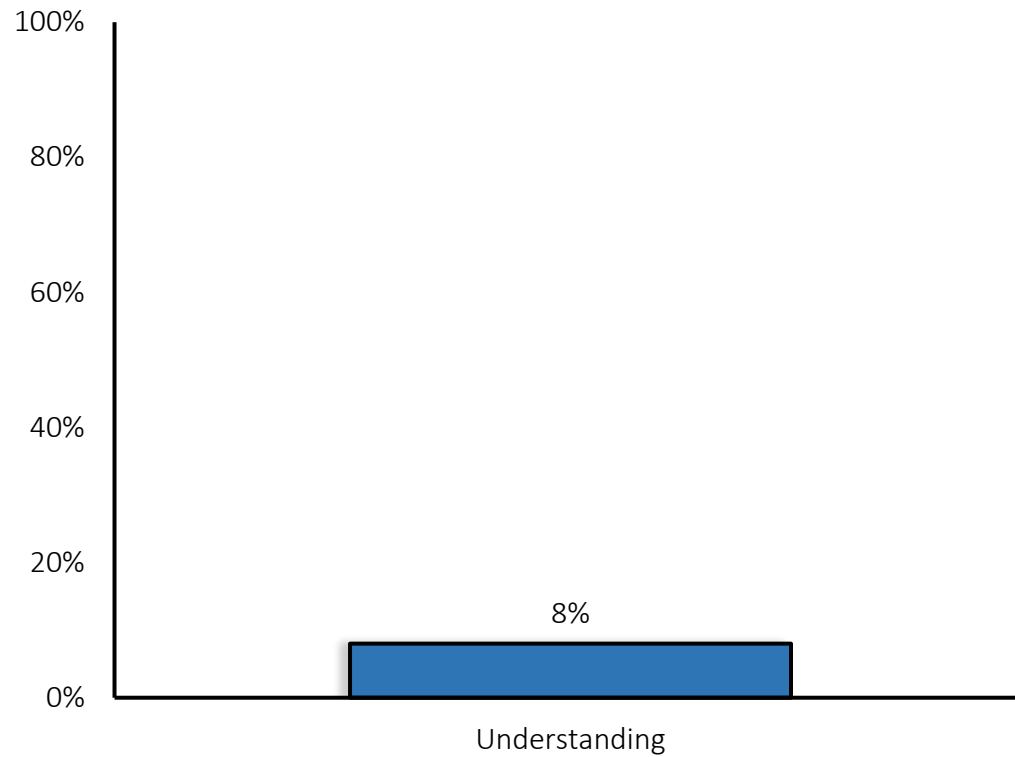


Question 8 - Haber

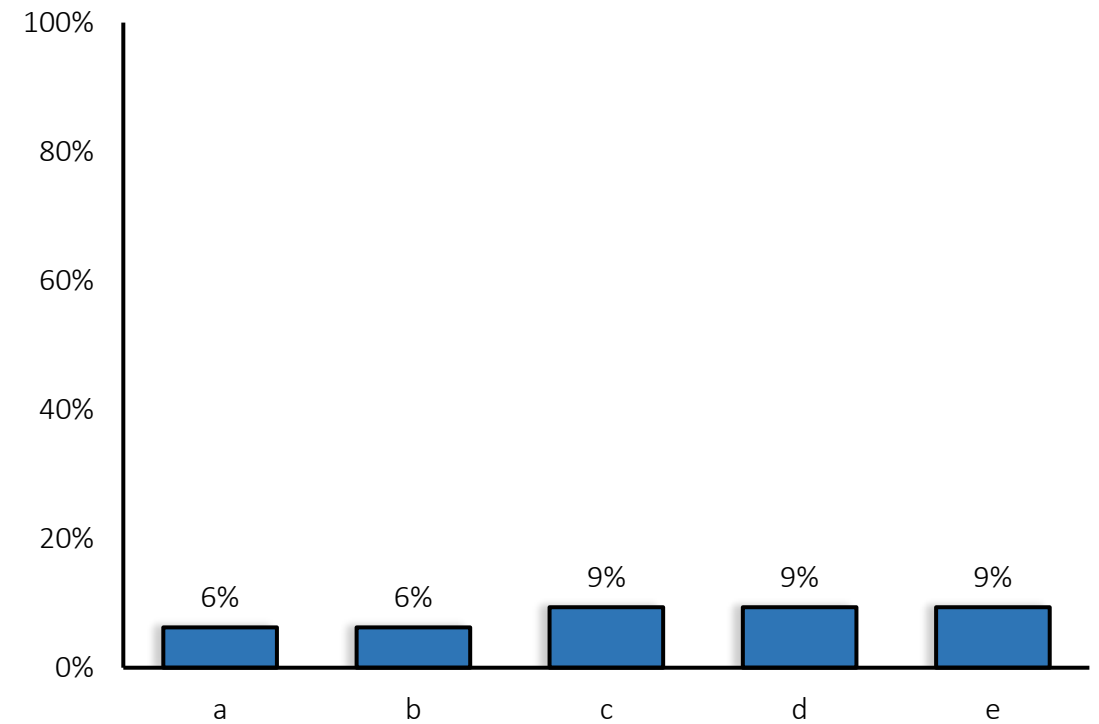


- Understanding was the only taxonomy in the question

Question 8 (Haber) Avg Score by Taxonomy



Question 8 (Haber) Avg Score by Sub-Question





Report



Performance Report

- Titration & Gravimetric Analysis

- Application-type questions averaged 12%, while Analyzation-type questions averaged 2%
- Part *a.ii – Defining Equivalence Point* & part *c.iii – Calculating Moles* received zeros for all students, while part *c.iv – Calculating Concentration* had 25% of students receiving full marks
- Bottom 5 students received zeros on all parts except part *e.i - Calculating Concentration* & part *e.iii - ****, while top 5 students averaged above 30% on parts *c.iv – Calculating Concentration*, part *d – Making a Standard Solution* & part *e.ii – Calculating Molar Absorptivity*

- Chromatography

- Knowledge-type questions averaged 17%, while Analyzation-type questions averaged 14%
- Part *a.i – Retention Time*, part *b – Chemical Principle of gL* & part *c.ii – Interpretation of a Chromatogram* averaged below 10%, while 56% of students received full marks on part *c.i – Interpretation of a Chromatogram*
- Bottom 5 students received zeros on all parts except part *a.ii – Mobile Phase* & part *c.i - Interpretation of a Chromatogram*, while top 5 students averaged above 40% on all parts except part *a.i – Retention Time* & part *b – Chemical Principle of gL*

- Hydrocarbon

- Understanding & Creation-type questions averaged 59%, while Analyzation-type questions averaged 64%
- Part *a – Identification of Functional Groups* received the highest average score of 88%, while part *c.iiia* & *c.iiib – Steps For Bond Breaking* averaged the lowest score of 25% & 34%, respectively.
- Bottom 5 students received zeros on part *c.iiia* & *c.iiib – Steps For Bond Breaking*, while top 5 students averaged above 60% on all parts



Performance Report

- Isomerism/EF
 - Application-type questions averaged 2%, while Knowledge-type questions averaged 77%
 - Part *b* – *Calculating Molecular Isomerism* averaged 2%, while the other two parts averaged above 50%
 - Bottom 5 students received zeros for part *b* – *Calculating Molecular Isomerism* & averaged below 20% for the other parts, while top 5 students averaged 5% on parts *b* – *Calculating Concentration* and averaged above 80% on the other 2 parts
- Reactions of Benzene
 - Knowledge-type questions averaged 17%, while Creation-type questions averaged 0%
 - Part *a.ii* – *Formation of Nitronium Ion* & part *c* – Reaction of Nitro Benzene of HCl averaged 0%, while 34% of students received full marks on part *b.ii* – *Nitration of Benzene*
 - Bottom 5 students received zeros on all parts of Question 5, while top 5 students received zeros on all parts except part *b.i* – *Nitration of Benzene* & part *b.ii* – *Nitration of Benzene*
- Acid/Base Prop of Hydrocarbons
 - Understanding-type questions averaged 0%, while Application-type questions averaged 28%
 - All parts of Question 6 averaged below 30%, with part *a.iii* - *pKa of Ethanoic Acid vs Ethanol* receiving an average score of zero. However, part *a.iii* had a 94% non-response rate.
 - Bottom 5 students received zeros on all parts except *b.i* – *Displayed Formula of Threonine*, while top 5 students averaged above 30% on parts *b.i* & *b.ii*



Performance Report

- Nitrogen Cycle
 - Evaluation-type questions averaged 35%, while Creation-type questions averaged 5%
 - Part *b – Prediction of Acid Rain*, part *e.iii – Bauxite Production* & part *e.iv – Bauxite Production* averaged below 20%, while 78% of students received full marks on part *e.i – Bauxite Production*
 - Bottom 5 students received zeros on part *b – Prediction of Acid Rain* & part *e.iv – Bauxite Production*, while the top 5 students averaged below 30% on parts *b – Prediction of Acid Rain*, part *d – Reasons for Government Reluctance* & part *e.iv – Bauxite Production*
- Haber Process
 - Question 8 tested students' understanding of the Haber Process, which averaged 8%
 - All parts (*a – e*) averaged below 10% with over 50% of students not responding to this question
 - Bottom 5 students received zeros on all parts, while top 5 students averaged below 20% on all parts except part *e*